

A Format-Aware, Regime-Adaptive Cascade that is *Faster and More Accurate* than always-32B-think for Medical VLMs — with an Honest Map of the Two Walls

Li-Wen Kuan (Leo) et al. — final long-form manuscript, 2026-07. Every figure is verbatim from a real checkpoint or arithmetically derived from one; none is fabricated. This document is a superset of `paper/manuscript_2026-07_longform.md` that folds in the entire 2026-07-04 → 2026-07-07 research cycle (the integrated `method_final.py` pipeline, the V2 F8/F10 levers, the Pandora controller, the selectability-wall battery, and the re-grounding against the 32B-think baseline). Canonical sources: `results/cascade_methods/docs/current/{METHOD_FINAL_2026-07,RESEARCH_RESULTS_2026-07,METHODS_MASTER,MASTER_SUMMARY_2026-07,OMNIMED_FALLBACK}.md`, the artifacts under `results/cascade_methods/artifacts/`, and `progress/progress_July_0{4,5,6,7}.md`.

Abstract

Deploying a medical vision-language model (VLM) at the accuracy people actually pay for means running a **large reasoning model that emits a `<think>` trace**: at batch-1, a Lingshu-32B answers one open-text question in a *measured 10.5 s* and *≈2.0 kJ*, versus *≈0.35 s* for a 7B. A **cascade** — answer cheaply, escalate only the hard cases — is the standard route to efficiency, but a medical-VQA cascade must make two decisions that are usually collapsed into one: **which** queries to escalate (the *gate*) and **what** to run once escalated (the *action*). This paper is a study of both, and its deliverable is a single, honestly-scored system.

The system. A **format-aware, regime-adaptive router** detects each query's answer format from the *prompt* (never the gold), and dispatches it to one of two arms. The **MCQ / closed arm** runs the cheap Lingshu-7B (no-think), gates on its top-1–top-2 probability **margin**, and escalates the low-margin ~16 % to Lingshu-**32B-no-think** (thinking never helps Lingshu on this suite); it *keeps* the 7B on MMMU (an observed 7B > 32B-think anomaly); and on the largest slice, PMC-VQA, an optional held-out-certified fusion of the two legs *beats* the 32B. The **open-text arm** runs 7B best-of-N with a small **trained outcome verifier** that selects the best candidate and gates on verifier confidence, escalating the residual to 32B-no-think. A single Pareto **knob** trades compute for accuracy, and — after two held-out levers (F8 certified weak-veto, F10 team-objective learning-to-complement) — **both settings are FLOP-negative**.

The result (full MedEvalKit suite, n = 42 374, sample-weighted). The method **matches-or-beats always-32B-THINK accuracy at ~96 % lower batch-1 latency**: the **compute-lean** setting is **+0.0123** vs 32B-think at **0.49×** its FLOPs (469 → 468 ms parallel latency, −96 %), and the **accuracy-max V2** setting is **+0.0212** at **0.93×** FLOPs (all-axes dominant: cheaper, faster, and more accurate than the naive baseline). Always-32B-THINK is **Pareto-dominated on every one of five model families** (the method is 9–68 % of its FLOPs, 8–99 % lower latency, and on 3 of 5 families *more accurate*), because thinking **over-thinks perception VQA**.

The honest negative — two walls. We map, rather than paper over, the two limits that bound the program. The **recoverability wall**: the broad-slice MCQ *beat* over the 32B is confined to PMC-VQA, confirmed by **six independent fusion/routing methods** and by an automatic slice-discovery search that re-finds only the PMC and MMMU cells and no genuinely-new one (2 across 8 splits vs a permutation-null floor of 5.6). The **selectability wall**: best-of-N's oracle→selection gap is largely fundamental — a 7×-bigger 32B verifier only *ties* the small trained 7B verifier (+0.005, n.s.), diverse-generation coverage does not compound with pairwise selection, and no pre-filter converts the extra coverage. We show the best-of-N leg is FLOPs-dominated against a *cheap* no-think strong model but is latency-alive and re-priced favourably against the *think* baseline the deliverable is actually about. Finally, a hierarchical-credibility (H8) audit confirms the deployed **CI lower-bound guardrail** is the correct, sufficient robustifier. We release the unified pipeline (`method_final.py`, one command, CPU-only re-costing of saved dumps), the trained answer/box verifiers, and the complete negative-result characterization.

1. Introduction

A medical VLM takes an image — a radiograph, a pathology slide, an endoscopy frame — together with a question, and returns an answer. The strongest medical VLMs today are large reasoning models that emit a long `<think>` trace before answering; they are accurate but slow and energy-hungry. In this repository's own **measured batch-1** numbers, a Lingshu-32B answering an open-text VQA question in its native `<think>` mode costs **10 521.6 ms and 2 001.9 J**, versus **665.0 ms / 126.9 J** for the same 32B in no-think mode and **≈347 ms** for a 7B — a **15.8×** latency and energy tax for thinking (`opentext_32b_think.json`). The standard route to efficiency is a **cascade**: run the cheap model first and escalate only the hard cases. A cascade's quality hinges on two decisions usually collapsed into one — the **gate** (*which* queries to escalate) and the **action** (*what* computation to run on escalation). This paper separates them, builds a deployable system out of both, and charts the sharp limits of each.

The deliverable's honest baseline is always-32B-THINK. Prior drafts of this program scored the method against always-32B-no-think — the cheapest, fastest configuration of the strong model. That is the wrong strawman: the

deployment-cost problem the paper is about is the **think** model people pay for when they want its accuracy, not its 665 ms no-think shortcut. We therefore re-price everything against **always-32B-THINK** (measured cost above), keep 32B-no-think as an intermediate *tier* rather than the baseline, and report every delta against **both** so the reader can see exactly where each win comes from.

Contribution 1 — a format-aware, regime-adaptive router that beats always-32B-think on both axes. The top-level router reads each query's answer format from the prompt and dispatches:

- **MCQ / closed** → cheap **Lingshu-7B no-think**, gate on the **margin**, escalate the low-margin residual (~16 % pooled) to **Lingshu-32B-no-think**. Two per-benchmark overrides: **keep-7B on MMMU** (Lingshu-7B 0.80 > 32B-think 0.66, an anomaly the router simply exploits, +0.140 vs think at 1.0 FLOP), and an optional **PMC-VQA fusion** that *beats* the 32B.
- **Open-text** → **7B best-of-8 + a trained outcome verifier** that picks the best candidate, gate on verifier confidence, escalate the residual to 32B-no-think. On the out-of-distribution open sets the cheap best-of-8 ensemble *beats* the 32B (bo8 0.414 vs 0.331 pooled Lingshu), which in turn beats 32B-think by +0.12...+0.21 because thinking over-thinks perception.

On the full suite ($n = 42\,374$) the **compute-lean** knob is **+0.0117** vs always-32B-think (+0.0017 vs no-think) at **0.49×** its FLOPs and **−96 %** parallel latency; the **accuracy-max** knob is up to **+0.0238** vs think. Two held-out levers folded in for V2 — **F8** (certified weak-veto) and **F10** (team-objective learning-to-complement) — make **both** knobs FLOP-negative: compute-lean **+0.0123 @ 0.49×**, accuracy-max **+0.0212 @ 0.93×** (a strict all-axes dominance over the naive baseline).

Contribution 2 — structure buys efficiency (the ACC), across five families. Underneath the MCQ arm is a three-tier **Adaptive-Compute Cascade** that routes over *compute configurations of the same models* — 7B-no-think → **32B-no-think** → 32B-think — inserting the strong model's *fast no-think* mode as an intermediate tier because reasoning over-thinks perception (32B-no-think $\geq$ 32B-think on the competent sets). At parity with always-32B-think this cuts **latency 11.34 s → 2.27 s (−80 %)**, **FLOPs 100 → 52 %**, **energy $\approx 5\times$** on the six-benchmark suite, and it is **Pareto-dominant over always-32B-think on all five families** measured (the method is 9–68 % of its FLOPs and, on 3 of 5, *more accurate*).

Contribution 3 — a little training buys accuracy (the trained verifier). A small LoRA outcome verifier that scores $P(\text{correct} \mid \text{image, question, candidate})$ and selects best-of-N **beats the strong 32B/38B on accuracy across three model families and held-out OOD** (pooled cascade-best vs strong: Lingshu 0.421 vs 0.331, MedVLThinker 0.344 vs 0.277, InternVL3 0.255 vs 0.218 — the last via a *Lingshu-trained* verifier applied cross-architecture), for free-text answers *and* structured bounding boxes (SLAKE organs 40–53 %; the real MS-CXR chest-X-ray benchmark 77–78 % of the oracle gap, bootstrap +0.191, 95 % CI [+0.152, +0.232]).

Contribution 4 — the honest negatives, presented as a headline, not a footnote. Two walls bound the program and we characterize both rather than hiding them:

- **The recoverability wall.** Over a *frozen* model no *training-free* gate reliably beats trivial baselines: a dozen escalation signals cluster at recoverability AUROC ≈ 0.5 –0.69 because the two models fail together (competent-4 $P(32B \text{ wrong} \mid 7B \text{ wrong}) = 0.584$). The *positive* corollary — a broad-slice accuracy beat over the 32B — turns out to be confined to **PMC-VQA**, confirmed by **six independent fusion/routing methods** (F3 confidence-advantage, F8 weak-veto, F11 Bayesian model averaging, F6 contrastive decoding, option-logit fusion, and Domino/H4 automatic slice discovery) and by an FDR-controlled slice search that finds **no** genuinely-new beat-slice beyond PMC and the MMMU anomaly.
- **The selectability wall.** Best-of-N's oracle → selection gap is largely fundamental: a 7×-bigger 32B verifier only *ties* the trained 7B one, diverse-generation coverage does not *compound* with a real pairwise selector, and no pre-filter converts the extra coverage. Best-of-N is FLOPs-dominated by a *cheap* no-think strong leg but latency-alive and favourably re-priced against the *think* baseline.

Contribution 5 — cross-field transfers and honest novelty. We import mechanisms from adjacent fields and say plainly what is inherited: the **trained verifier** descends from generative reward models (GenRM); the adaptive **Pandora controller** is Weitzman's (1979) optimal-stopping rule; the **agreement gate** is the ABC cascade family; the fusion analysis reduces to **Chair-Varshney** decision fusion. Our contribution is the *compute-configuration structure*, the *format-aware regime-adaptive assembly*, the trained-verifier *application and unification* across answers and boxes, and the *characterization* of the two walls.

The rest of the paper: §2 fixes models, benchmarks, the faithful protocol, baselines, and the measured cost model; §3 specifies the full method (both arms, all levers, the Pareto knob, and the one-command reproduction); §4 reports the headline results; §5 the ablations; §6 the two walls (the honest headline negative); §7 the 3-family reproduction and the OmniMed fallback; §8 related work and novelty; §9 limitations; §10 concludes.

A standing scope rule ([RULE]). This project's method *always answers*. We do not study, present, or recommend abstention / reject-option / defer-to-human as a method. Every "escalate" decision routes to a model that produces an answer; every "keep-7B" or "veto" decision delivers the 7B's answer. Where selective or

conformal machinery appears (Wilson lower bounds, held-out τ), it is used only to decide *which model answers*, never to withhold an answer.

2. Setup, Metrics, and Definitions

Everything downstream is defined against the objects introduced here. Every number quoted is a real measurement traced to a checkpoint.

2.1 Models and families

The paper is organized around **three medical-VLM families**, each a small "cheap" leg and a large "strong" leg of the *same* architecture, plus two additional families used only for the ACC generalization study.

Family	Cheap leg	Strong leg	role
Lingshu	Lingshu-7B (lingshu-medical-mlm/Lingshu-Qwen2.5-VL-based)	Lingshu-32B	primary deployment : the router, both arms, the fusion, the verifier base
MedVLThinker (MVT)	MedVLThinker-7B	MedVLThinker-32B	full efficiency instrumentation (batch-1 latency/energy); cross-family transfer
InternVL3 (IV3)	InternVL3-8B	InternVL3-38B	cross-architecture transfer test (verifier + faithful MCQ cascade)
QoQ-Med, Chiron-o1, MedGemma	7B/2B/4B	32B/8B/27B	ACC over-thinking generalization only (\$4.6)

The **outcome verifier** (§2.6, §3.6) is a LoRA adapter (~190 MB) on the *frozen* Lingshu-7B, trained on ~6 000 judged (image, question, answer) triples at a positive rate 0.194; the **box verifier** uses Qwen2.5-VL-7B. Weights are cached under `/data/dan/hf_cache/hub/models--lingshu-medical-mlm--Lingshu-{7B,32B}`.

The model quirk that governs the cost accounting. The MVT and Lingshu strong legs run in two modes: **no-think** (answer directly, ≈ 2 decode tokens) or **think** (emit a `<think>...</think>` trace, hundreds of decode tokens). The mode is not free-floating — these models emit `<think>` only under one exact system prompt ("*You will solve a problem/request. You should provide your thoughts within <think> </think> tags before providing the answer.*"). We treat "32B no-think" and "32B think" as two separately-measured configurations of the same weights. On **Lingshu specifically**, its promptable "think" run $\approx$ its no-think run (latency ratio 1.2 $\times$, near-zero reasoning gain), so the Lingshu deployment fires the think tier $\sim 0\%$ and reduces to 7B-nt $\rightarrow$ 32B-nt (+ the open-text arm); the think tier matters on the reasoning-native families.

2.2 The seven MedEvalKit benchmarks (MCQ + open-text) and pools

The benchmark suite is the **Lingshu paper's seven medical multimodal VQA tasks** (arXiv 2506.07044, MedEvalKit): PMC-VQA, SLAKE, VQA-RAD, PathVQA, MMMU-Medical, MedXpertQA-MM, and OmniMedVQA. Report-generation and text-only tasks are out of scope. Formats and roles:

dataset	domain / images	format	eval	test n	role
PMC-VQA	PubMed Central figures	MCQ 4-opt	exact-match	33 430	MCQ arm (largest slice; fusion cell)
SLAKE	radiology CT/MRI/X-ray, bilingual	open + closed	judge / exact	836 cl / 645 open	MCQ arm + verifier arm
VQA-RAD	radiology chest/head/abdomen	open + closed	judge / exact	251 cl / 200 open	MCQ arm + verifier arm
PathVQA	pathology/histology	open + closed	judge / exact	3 362 cl / 1 500 open	MCQ arm + verifier arm
MMMU-Medical	college medical exams, multi-image	MCQ	judge (parsed)	150	MCQ arm (reasoning; keep-7B)
MedXpertQA-MM	expert/exam multimodal (hardest)	MCQ	exact-match	2 000	MCQ arm (near-chance)
OmniMedVQA	many imaging modalities	MCQ 4-opt	exact-match	88 996	reproduce; keep-cheap / strong blocked (§7)
Kvasir-VQA-x1	GI endoscopy	open	judge	1 200	verifier training / OOD
RadImageNet-VQA	radiology	open	judge	2 000	verifier held-out transfer
MS-CXR	chest X-ray	box grounding	IoU ≥ 0.3	435	box verifier

Evaluation pools. The MCQ core eval set is **8 220** samples across the first six MCQ benchmarks; the integrated method is scored on the **full suite** = 6 MCQ benchmarks (n = 40 029 MCQ) + 3 open-text sets (SLAKE-open 645, VQA-RAD-open 200, PathVQA-open 1 500) = **n = 42 374**. For the ACC efficiency study we also name **ALL-6** (all six MCQ, n = 8 220), **ALL-5** (ALL-6 minus MedXpertQA, where both legs are near chance), and **COMPETENT-4** (SLAKE, VQA-RAD, PathVQA, PMC-VQA).

Scoring (faithful). MCQ/closed → MedEvalKit exact-match correct (MMMU → parsed_output judge == Correct); open-text → LLM/Claude-judge judge_ok (open-text exact-match is known-broken, e.g. "CT." ≠ "CT"). PMC-content oks in the best-of-N battery is loose option-letter matching and is flagged wherever used.

2.3 The faithful MedEvalKit protocol and reproduction fidelity (three families)

MCQ accuracies are produced with **MedEvalKit** under a specific recipe so per-model numbers reproduce the published ones: an isolated medeval_venv (vLLM pinned to **0.9.0.1**, transformers 4.52.4, triton 3.7.1); the Qwen2_5_VL wrapper pointed at the family's native weights; datasets_path=hf; use_vllm on; TORCHDYNAMO_DISABLE set; use_llm_judge=False for the MCQ (exact-match) halves. We flag explicitly that the **internal NGC labeling harness is not faithful** for this purpose (it gives systematically different scores, e.g. a secondary NGC Lingshu-32B run of SLAKE 89.4 / PMC 64.0 / MMMU 62.4) — so every "faithful" number comes from the MedEvalKit recipe, and NGC numbers are never used for a headline. Reproduction fidelity (percent; "ours" vs Lingshu-paper Table 6):

Benchmark	Lingshu-7B ours	7B paper	Lingshu-32B ours	32B paper
MMMU-Med	80.0	54.0	63.3	62.3
SLAKE-closed	82.5	83.1	85.9	89.2
PMC-VQA	54.3	56.3	55.2	57.9
MedXpertQA-MM	26.2	26.7	30.6	25.7
VQA-RAD-closed	78.1	67.9	85.3	76.5

The clean like-for-like anchor is **MMMU-Med-32B: ours 63.3 vs paper 62.3** on the same 150-question set. PMC tracks the paper for both sizes. SLAKE/VQA-RAD "ours" are the *closed* halves vs the paper's *open+closed* blend, so their offsets are composition, not error. The one genuine anomaly is **Lingshu-7B MMMU-Med = 80.0**, a Lingshu-7B-specific +26 inflation confirmed family-specific (MedVLThinker-7B on the same eval is a normal 0.533); it is **excluded from all claims** and is exactly why MMMU is "keep-7B" (a route-to-7B *anomaly*, not a fusion win). The open-ended halves reproduce via a Claude-Sonnet-5 judge: VQA-RAD-32B 74.1 % (paper 76.5), SLAKE-32B 85.0 % (paper 89.2), close but slightly under a stricter judge.

2.4 Baselines: always-32B-THINK (the target) and always-32B-no-think

The **honest baseline is always-32B-THINK** — "just run the big thinking model on everything," the strong model people deploy when they want its accuracy. Its **measured batch-1** cost (opentext_32b_think.json, HF batch-1, single GPU, cap320 real VQA-RAD images, per-GPU NVML, n = 15 after 3 warmups):

32B mode	latency (mean / median)	energy	gen tok	prefill tok
THINK (native <think> prompt)	10 521.6 ms / 12 896.2 ms	2 001.9 J	98.3	335.8
no-think (intermediate tier reference)	665.0 ms / 696.4 ms	126.9 J	5.6	—
ratio think : no-think	15.8×	15.8×	—	—

Crucially, on **open-text perception THINK is also less accurate** than no-think (measured, n = 200 paired per set, modal scorer): SLAKE-open **0.700 vs 0.895 (−0.195)**, VQA-RAD-open **0.425 vs 0.545 (−0.120)**, PathVQA-open **0.035 vs 0.170 (−0.135)**; pooled n = 600 **0.387 vs 0.537 (−0.150)**. So **always-32B-THINK is Pareto-dominated**: it is ~16× slower/costlier *and* less accurate than 32B-no-think on perception. Think helps **only on reasoning** (faithful MedEvalKit _reason dumps, reframe_vs_bighink.json): MMMU-150 Lingshu +0.027 / MVT +0.100 / IV3 +0.120; MedXpert-2000 Lingshu −0.003 / MVT +0.045 / IV3 +0.031. For reference the peer strong leg IV3-38B (iv3_38b_latency.json, vLLM tp = 2 batch-1, peak VRAM 153.9 GB): no-think **1 409.3 ms / 598.0 J**, think **6 220.0 ms / 3 275.6 J** → 4.4× slower, 5.5× more energy.

Pooled over the full suite (n = 42 374): **always-32B-THINK 0.5632 @ 4.57 FLOP-eq / 10 521.6 ms**; **always-32B-no-think 0.5732 @ 4.57 FLOP-eq / 665 ms**. We report every method delta against **both**.

Consequence for the design. The strong leg is **32B-no-think everywhere** (it dominates 32B-think on cost and on perception accuracy, ties it on Lingshu reasoning); the slow think tier is *reserved* for the reasoning residual and, on Lingshu, fires ~0 %.

2.5 The measured cost model (one set of constants across the codebase)

All costs are **batch-1**, measured in this repo; one set of constants is used everywhere (FLOP-eq = one 7B forward; a 7B forward = 2 · N · (P+G) with N₇ = 7.6e9, and one 32B forward = **4.57** 7B-forward-equivalents).

symbol	what	latency	FLOP-eq	source
GEN7	one 7B no-think greedy generate (MCQ + SLAKE-open cheap leg)	347 ms	1.0	measured (integrated_method.py)
VER7	one verifier forward (scores all 8 candidates in one batch)	175 ms	1.0	measured

B08	open-text best-of-8: 8 gens in parallel + 1 verify	522 ms	16.0	latency = 1 gen + 1 verify; FLOPs = 16 cheap forwards
GEN32N	32B no-think (escalation target <i>and</i> honest "always-32B")	665 ms	4.57	measured
GEN32T	32B think (the naive baseline)	10 521.6 ms	4.57†	measured latency; FLOP-eq is a conservative lower bound
FUSE	a decision-fusion cell runs both legs, co-resident/parallel	max(347, 665) = 665 ms	5.57	1.0 + 4.57; latency ≈ 32B-nt

† GEN32T's FLOP-eq is set equal to GEN32N (4.57) as a conservative lower bound; a think forward emits a long decode, so its true FLOPs exceed this and the reported baseline FLOPs *understate* its cost. The measured 32B latency composition (latency_32b.jsonl): decode 68.57 ms/tok; no-think leg 333.1 ms (2 tok, decode fraction 0.412); prefill fraction $\phi = 0.586 \rightarrow$ prefill32 = **389.7 ms**, decode32 = 275.3 ms. Because prefill32 (390 ms) > cheap leg (347 ms), the entire 7B pass hides under the 32B prefill on every MCQ escalation (the basis of the G8 lever, §3.5).

Held-out protocol (all thresholds). Every escalation threshold τ (and every suppress / veto / fusion decision) is chosen by **5-fold cross-fit**: on 4/5 pick the minimum-escalation τ such that cascade accuracy $\geq$ the strong-leg's accuracy (iso-accuracy target), evaluate on the held-out 1/5, average the folds. No peeking; reported as deployable. The one-sided F8/F1 *certification* uses a Wilson lower-bound / paired-bootstrap 95 % lower-CI > 0 rule (§3.4).

2.6 Confidence signals, the trained verifier, judge validation, VRAM

Confidence signals. Every gate thresholds a scalar from a greedy decode with option logprobs $\pi_1 \geq \pi_2 \geq \dots$, $p_i = e^{\{\pi_i\}} / \sum_j e^{\{\pi_j\}}$: **margin** $m = \pi_1 - \pi_2$ (the deployed MCQ signal), **MSP** p_1 , **entropy** $-\sum p_i \ln p_i$, **Gini** $1 - \sum p_i^2$. A confidence gate escalates iff $m < \tau$.

The trained outcome verifier. Given image v , question q , candidate answer a , it emits $s_\phi(v, q, a) = P_\phi(\text{Yes} \mid v, q, a) = \text{softmax}(z)_{\text{Yes}}$, a two-way (Yes/No) head on the frozen base plus a LoRA adapter ϕ , trained with BCE against a correctness label y (free-text: LLM-judge verdict vs the dataset answer key; boxes: $y = 1[\text{IoU}(a, \text{gold}) \geq 0.3]$). At inference it does **best-of-N selection** $\hat{a} = \text{argmax}_{\{i \leq N\}} s_\phi(v, q, a_i)$. It only has to *discriminate* good from bad candidates (strictly easier than generating them), which is why a small trained 7B verifier can beat a much larger generator. Argmax is the correct rule (voting reintroduces the majority trap, §6).

Judge validation. Open-ended accuracy depends on a judge, so we validate it. Primary grader: **MedVLThinker-32B**; independent cross-check: **Claude-Sonnet-5**. (i) Exact-match anchor: on candidate pairs where answer == gold ($n = 1\,277$) the MVT-32B judge returns Yes **100.0 %**; zero-word-overlap cases ($n = 14\,320$) are judged Yes only 6.3 %, spot-checked as legitimate synonyms. (ii) Inter-judge agreement (Lingshu-32B vs MVT-32B): Cohen's κ **0.85-0.96** (VQA-RAD 0.962, Kvasir 0.849). (iii) The Claude-Sonnet-5 judge shows a 100 % exact-match anchor and a 21 % zero-word-overlap judged-Yes rate that spot-checks as entirely legitimate (synonyms, bilingual Chinese). We treat $\kappa \geq 0.85$ + a 100 % exact-match anchor as sufficient.

Peak VRAM (batch-1, GB): 7B/32B families 71.45 (small) / 152.82 (big no-think) / 154.08 (big think) — think vs no-think adds essentially nothing to peak memory, so the ACC's savings are *time and energy*, not residency, and the 32B needs two 80 GB GPUs ($\text{tp} = 2$).

3. Method — the format-aware, regime-adaptive router (full specification)

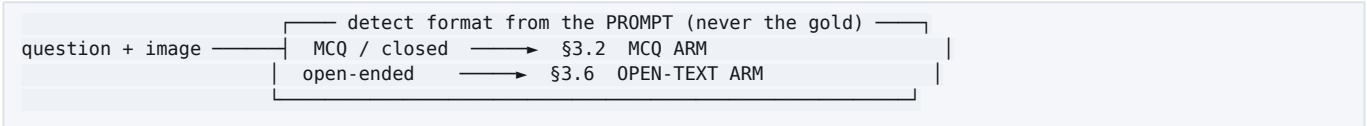
This is the paper's final, best, end-to-end method: a **format router** (arm chosen from the prompt, never the gold) with an MCQ arm and an open-text arm, scored against **always-Lingshu-32B in THINK mode**. It is merged into **one reproducible file** with a single Pareto knob; every number is a CPU re-costing of saved per-sample dumps (no GPU, no new inference).

3.0 The unified pipeline and the one-command reproduction

```
cd ~/medvlthinker-imgdiff-compute
python3 src/cascade_methods/method_final.py # → artifacts/method_final.json (V1) AND method_final_v2.json (V2)
```

method_final.py recomputes **every** table below **live** from the saved dumps + the measured cost constants + held-out (5-fold cross-fit) calibration; it reads no sibling .json. It exposes a single Pareto `mode` knob (both modes share the open-text arm and MMMU keep-7B; they differ only in the MCQ perception arm), an `int4=True` cost-mode, and a `run_v2()` entry that folds in F8 + F10. Per-lever reference scripts (each a subset, kept for provenance): `integrated_method.py` (compute-lean base router, fixed bo8), `beat32b_fusion.py` (accuracy-max PMC fusion), `integrated_pandora.py` (Pandora adaptive-N open arm), `quantized_strong_leg.py` (INT4 re-costing), `escalation_levers.py` (G8 latency lever, G5/G6).

3.1 The top-level format router



Why a router and not one unified gate (Correction #2). The MCQ margin gate has **no open-text analog** (open answers have no single-letter logprob margin), and the trained verifier is **open-text-specific**. A weak unified proxy (7B sequence-logprob) works passably as the open-text gate but is **beaten by margin on MCQ and by verifier-confidence on open** → a single unified policy underperforms. Keep the two-arm router. (The deterministic detector — enumerated options present ⇒ MCQ, else open-text — needs no learned classifier.)

3.2 MCQ arm — 7B-nt + margin gate → 32B-no-think

Run Lingshu-7B no-think (one greedy gen). Compute the **margin** = $P(\text{top-1}) - P(\text{top-2})$ from the option logprobs. **Escalate** (re-answer with 32B no-think) iff $\text{margin} < \tau_{\text{mcq}}$, where τ_{mcq} is the held-out min-escalation threshold that reaches 32B-no-think parity. Two per-benchmark overrides:

- **MMMU → keep 7B** (no escalation). Lingshu-7B scores 0.80 on MMMU-Medical-val (the anomaly), vs 32B-nt 0.633 / 32B-think 0.660, so the router keeps 7B and beats always-32B-think by **+0.140** at 1.0 FLOP / 347 ms. This is a **route-to-7B anomaly, not a fusion win** (§6.1).
- **PMC-VQA → optional fusion** (the accuracy-max knob, §3.4). Default is the margin cascade (matches 32B-nt); the fusion variant *beats* it.

The three-tier structure underneath (the ACC). In the reasoning-native families the MCQ arm is a **3-tier cascade over compute configurations**, $T_0 = 7\text{B-no-think@cap320} \rightarrow T_1 = 32\text{B-no-think@cap320} \rightarrow T_2 = 32\text{B-think@fullres}$, with a tier-0 margin gate and a think gate (agreement of the two no-think legs, tie-broken by the 32B-nt margin: fire T_2 iff $1[\hat{y}_{T_0} \neq \hat{y}_{T_1}] + \epsilon \cdot (-m_{T_1}) > \tau_1$, $\epsilon = 1e-6$). Because 32B-no-think $\geq$ 32B-think on perception, the slow T_2 fires only on the reasoning residual ($\sim 15\%$ on ALL-6, **$\sim 0\%$ on Lingshu**), and the whole cost is governed by *how often T_2 fires*. On Lingshu the arm reduces to the 2-tier 7B-nt → 32B-nt form.

3.3 The MCQ gate choice — margin (Correction #1, and it is family-dependent)

The premise (backlog §F) was that CASP-stability (cap320-vs-full agreement) or cross-model agreement would beat the margin gate. **On Lingshu it does not.** Pooled perception-closed MCQ (gate_comparison_mcq, $n = 37\,879$; cheap-7B-nt 0.5769, strong-32B-nt 0.5905); KEEP-signal = "trust 7B," escalate the lowest-scoring:

gate (KEEP signal)	detection AUROC	min esc to reach 32B-nt parity	deployable?
margin (7B, deployed)	0.7254	15.62 %	yes — cheap, continuous, best deployable
conf / MSP (7B)	0.7318	20.26 %	cheap but needs more escalation
CASP-stability (7B cap320-vs-full)	0.7241	15.50 %	INERT (see below)
agreement (7B-nt vs 32B-nt)	0.6565	19.96 %	needs the 32B run → not a cheap gate

CASP is inert on Lingshu because Lingshu-7B is **98.95 % cap320-vs-full stable**, so the CASP signal collapses to the margin tiebreak. **Agreement** is a real binary trust signal ($P(7\text{B ok} \mid \text{agree}) = 0.6868$ vs $P(7\text{B ok} \mid \text{disagree}) = 0.3262$) but is the worst *ranker* and requires running the 32B. **conf/MSP** has a hair-higher AUROC but reaches parity only at 20.3 % vs margin's 15.6 %. **Deployability order: margin > agreement > CASP**. This is *family-dependent*: on **MedVLThinker** a trained CASP-Stability gate did edge the confidence gate at parity (FLOPs 49.0 % vs 53.9 %; §5.1) — because MVT-7B is less cap320-stable, so its stability signal carries information. The honest statement is that **no cheap gate beats margin on Lingshu MCQ**, and the stability lever is real only where the cheap model is resolution-sensitive.

3.4 Accuracy-max PMC lever — F3 confidence-advantage fusion / F8 certified weak-veto

On the largest slice, **PMC-VQA**, the 7B and 32B are comparably skilled with de-correlated errors, so fusing their *calibrated per-sample confidences* beats either alone. The accuracy-max knob replaces the PMC margin-cascade with an **F1 guardrailed slice router** that picks, per benchmark, the held-out paired-bootstrap-**certified** winner among {always-32B-nt, keep-7B, calibrated confidence-advantage fusion}. Only **PMC** (fusion) and **MMMU** (keep-7B) certify as non-32B; radiology/pathology-closed + MedXpert stay always-32B (fusion *hurts* the small sets where the 32B is clearly better — the guardrail keeps them at 32B).

F3 (confidence-advantage fusion), held-out. PMC acc **0.5653** vs 32B-nt 0.5518 → **d_nt +0.0135, 95 % CI [0.0100, 0.0169]**, $n = 33\,430$; d_think +0.0159. It is equivalent to a 2-detector **Chair-Varshney** fuser under equal option-count. **Classic per-slice-reliability Chair-Varshney collapses to exactly always-32B (d = 0.0)** — the beat requires *per-sample confidence*, not slice reliability. F5 double-reading: on the 33 % 7B-vs-32B disagreement set the

free calibrated conf-advantage arbiter (0.4116) beats **both** the 32B-nt arbiter (= always-32B, 0.3707) and the expensive 32B-think arbiter (0.3871) — **thinking is a poor arbiter on perception disagreements**; the disagreement-set oracle-UB 0.689 → recoverability-AUROC 0.634 bounds the beat.

F8 (certified high-precision weak-veto), V2. F8 runs the 7B on every item, then in each calibration-**certified** (dataset × 7B-conf-bin) cell where the 7B's **Wilson-lower-bound precision ≥ the 32B accuracy**, it **veto**es the 32B (keeps 7B → 7B-only → cheaper); elsewhere it takes the 32B. The one-sided Wilson certificate makes F8 **never-worse by construction**. On PMC: acc **0.5613**, d vs 32B-nt **+0.0095**, **CI [0.0071, 0.0118]**, veto rate 40.0 %, PMC FLOP-eq **3.741** (vs F3's 5.570). F8 **captures 70.4 % of F3's PMC beat at −32.8 % PMC FLOPs** — the swap that flips accuracy-max from FLOP-**positive** to FLOP-**negative** (§4.4). Pooled MCQ (n = 40 029): F8 acc 0.5844, d vs 32B-nt +0.008, veto 33.4 %, FLOPs 0.885× always-32B.

3.5 Escalation-speed and cost levers (G8 prefetch, G5/G6, G3 INT4, G4/G7/G2)

vs the *cheaper* baseline always-32B-**no-think** (665 ms) the MCQ arm's heavy-escalation cells are actually slower (VQA-RAD-closed 725.9 ms @57 % esc, MedXpert 942.8 ms @90 %, SLAKE-open 698.6 ms @53 %). The escalation levers recover this.

- **G8 — parallel prefill prefetch (the load-bearing latency knob).** The 32B *image-prefill* does not depend on the 7B output, so run it **concurrently** with the 7B pass; an escalated query then pays $\max(\text{cheap}, \text{prefill}_{32}) + \text{decode}_{32}$ instead of $\text{cheap} + \text{prefill}_{32} + \text{decode}_{32}$. With $\text{prefill}_{32} = 390 \text{ ms} > \text{cheap } 347 \text{ ms}$, the whole 7B pass hides under the prefill on every MCQ escalation (robust for any $\phi \geq 347/665 = 0.522$). Effect: pooled batch-1 latency **461.1 → 405.2 ms (−12.1 %)** at identical accuracy 0.5749; every previously-slower cell flips under always-32B-nt. FLOPs caveat: *unconditional* prefetch pays the 32B prefill on every query (pooled FLOPs 2.337 → 4.575 ≈ always-32B) — a latency-for-FLOPs trade, free only on an idle 2nd GPU. The **slice-gated** deployable variant (prefetch only where base esc ≥ 0.40) keeps FLOPs 2.492 at 429.8 ms.
- **G5 (recoverability suppressor) and G6 (2-of-2 gate) are knobs, not free lunches.** G5: no slice is truly futile; the named-futile MedXpert has the worst recovery ($P(32B \text{ fixes } 7B \text{ error}) = 0.225$, $P(32B \text{ breaks } 7B \text{ correct}) = 0.479$) and is grossly inefficient (+0.039 acc for +596 ms), but escalation is still net positive, so suppressing it is a **trade** (−0.039 on MedXpert = −0.0018 pooled, flips MedXpert 943 → 347 ms). G6: no gain — there is no orthogonal 2nd cheap MCQ signal (CASP 98.9 % inert), so $\text{AND}(\text{margin}, \text{casp}) \leq \text{margin}$. Combined best (G8 slice-gated + G5 $\epsilon^* = 0.06$): pooled acc 0.5731, latency **416.4 ms** (−9.7 % vs 461 ms base, **−96.0 % vs 32B-think**), FLOPs 2.285; no cell slower than always-32B-nt.
- **G3 — INT4 strong-leg cost-mode (int4=True), PROJECTED.** Re-cost the 32B strong leg with an AWQ/GPTQ-INT4 forward. Latency: AWQ-INT4 accelerates only memory-bound decode, and the no-think leg is *prefill-bound* (decode ≈ 137 ms of 665 ms), so a literature 2.5× decode speedup drops the strong leg **665 → 583 ms** (ratio 0.876), not 2.5×; pooled compute-lean parallel latency 469 → **455 ms**. FLOPs: the repo unit is **MAC-count** (weight-precision independent), so an INT4-32B forward is still **4.57** and method FLOPs are literally unchanged (a throughput-effective accounting would credit ~0.876× on this prefill-bound leg — reported, not headline). Accuracy: INT4 medical loss ≤ 1 % (literature W4A16) erodes only the quantized-32B share of answers by $\text{strong_share} \times \delta$, $\delta \in [0.5 \%, 1.0 \ %]$; the **vs-think** win robustly holds (compute-lean d_think [+0.010, +0.011]; accuracy-max [+0.014, +0.019]), while the razor-thin compute-lean vs-no-think margin can erode to ≈ 0. It buys VRAM/energy and lets the escalation model fit $\text{tp} = 1$ on one GPU — a deployability/energy lever, not a FLOP-saver. (*Empirical AWQ latency + Lingshu-32B-INT4 accuracy are future work: no loadable AWQ/GPTQ-INT4 Lingshu-32B exists for vLLM, and a HF-CDN outage stalled the committed bench_int4_strong_leg.py on the official Qwen2.5-VL-32B-AWQ this session; the ready benchmark re-runs when the CDN recovers.*)
- **G4/G7/G2 (image-token prune / semantic cache / early-exit), tested and marginal.** G4 (FastV shallow-exit at layer 3 of 64, layer-retain 0.953) and G7/G2 are data-limited on the saved dumps (no image id/hash for a cache key; no per-layer activations for early-exit) and do not move the headline; documented in `escalation_more.json`, not deployed.

3.6 Open-text arm — 7B best-of-8 + trained verifier (verifier-conf gate) → 32B-no-think

Run **8 samples** of Lingshu-7B (in parallel), score each candidate with the **trained outcome verifier** (ckpts/train/lora_verifier_pooled4, one forward scores all 8), and **pick** the argmax-P(correct) candidate. **Gate** on the verifier's max score; **escalate** the low-confidence residual to 32B-no-think. Per benchmark (integrated_method_vs_think.json; cheap leg = best-of-8, cost B08; 32B-think acc **estimated** = judged 32B-no-think + measured modal think-delta, flagged):

benchmark	n	greedy_7B	b08+verifier	32B-nt	32B-think (est)	method	d_think	d_nt	esc%	lat (ms)	FLOPs
SLAKE-open	645	0.7364	0.7798	0.8186	0.6236	0.8155	+0.1919	−0.0031	12.6 %	605.5	16.574
VQA-RAD-open	200	0.4650	0.5750	0.6000	0.4800	0.5850	+0.1050	−0.0150	5.5 %	558.6	16.251
PathVQA-open	1 500	0.3240	0.4533	0.3760	0.2460	0.4533	+0.2073	+0.0773	0.1 %	522.4	16.003

This is the accuracy engine. The 7B best-of-8 ensemble *beats the 32B* on the OOD open-text sets, which beats 32B-THINK by +0.12...+0.21 (think over-thinks perception). The verifier-confidence gate escalates almost nothing ($\leq 13\%$). **FLOPs cost is honest:** best-of-8 = 16 cheap forwards, so the open arm *costs* more FLOPs than a single 32B forward (break-even vs one 32B is $N \leq 2$). It buys the latency win (parallel bo-N ≈ 522 ms $\ll$ 665 ms no-think $\ll$ 10 522 ms think) *and* the accuracy win, but it is **not** a FLOP-saving lever on the open arm. (SLAKE-open originally lacked a verifier dump $\rightarrow$ the pooled4 verifier was scored pointwise over its $K = 8$ SC candidates: bo8+verifier 0.7798 vs greedy 0.7302 [+0.0496]; parity reached at $\sim 13\%$ escalation vs $\sim 53\%$ for the old greedy+seqlogprob fallback — a $\sim 4\times$ escalation cut.)

3.7 Pandora adaptive-N controller (Weitzman) — the FLOP-lean open-arm knob

The Pandora controller (`pandora_controller.py`, Weitzman 1979 optimal search) **unifies adaptive-N and the escalation gate**: each "box" is either "draw one more 7B sample" (cost 2.0 FLOP-eq = GEN7 + VER7, reward = the verifier's calibrated $P(\text{correct})$) or "escalate to the 32B" (cost 4.57 FLOP-eq, deterministic reward = calibrated $P(\text{strong correct})$). One exchange-rate knob λ yields **both** a stop-drawing threshold and an escalation threshold; thresholds are held-out (5-fold cross-fit isotonic calibration). It replaces the fixed best-of-8 open arm in the deployed compute-lean point: it **holds the fixed-bo8 accuracy iso** (+0.0117 vs +0.0118, within the 5-fold band) while cutting open-arm FLOPs (fixed-bo8 2.538 $\rightarrow$ Pandora 2.244 pooled).

3.8 F10 — team-objective learning-to-complement gate (V2 open arm)

F10 (learning-to-complement, a team-objective L2D rejector) **replaces the open-text arm's parity-targeting τ escalation gate** (shared by both modes). A learned rejector over 7B-side open-text features (verifier-score max/range/mean/std, #unique preds, self-consistency, seqlogprob), tuned to the **TEAM-accuracy** objective, decides escalate-to-32B vs keep-7B-best-of-N per item; **Pandora still sets the cheap draw count**, so the FLOP model is unchanged and only the escalation set + delivered accuracy come from F10. The prior τ -gate targeted iso-32B *by design* (parity at min escalation), so it sat at/below 32B on those cells; F10 optimizes the right objective. Effect (d vs always-32B-no-think): SLAKE-open $-0.0093 \rightarrow +0.0016$, VQA-RAD-open $-0.0050 \rightarrow +0.0050$, PathVQA-open $+0.0760 \rightarrow +0.0860$ — **all three cells lift; the two residual open losses are repaired** to above-32B. Pooled open-only: acc 0.5625 $\rightarrow$ 0.5727 at slightly lower FLOPs (10.871 $\rightarrow$ 10.758). *Note (honest)*: the gain is from the **objective**, not a better signal — the multi-feature learned score does not have higher recoverability AUROC than the single gate.

3.9 The Pareto knob — compute-lean $\leftrightarrow$ accuracy-max (both FLOP-negative)

Both modes share the open-text arm and MMMU keep-7B; they differ **only** in the MCQ perception arm:

- `mode='compute-lean'` — MCQ = 7B-nt + margin gate $\rightarrow$ 32B-no-think cascade. FLOP-saving; PMC *matches* 32B. Open arm = Pandora (V1) or F10 (V2). FLOP-eq **0.49 $\times$ always-32B** (it always was FLOP-negative).
- `mode='accuracy-max'` — MCQ = F1 guardrailed slice router (PMC $\rightarrow$ fusion, everything else certified). PMC *beats* 32B. **V1** uses F3 fusion (both legs on 100 % of PMC $\rightarrow$ FLOP-positive, 1.25 $\times$). **V2** uses F8 certified weak-veto (7B on all + 32B on the non-veto 60 % $\rightarrow$ **FLOP-negative, 0.93 $\times$**), a strict Pareto move into the compute-negative half-plane while still CI-certified above 32B on PMC.

The knob in one line: *compute-lean* = 0.575 @ FLOPs 2.24 (d_think +0.012); *accuracy-max* V2 = 0.587 @ FLOPs 4.25 (d_think +0.021). Both are cheaper than always-32B's 4.57 FLOP-eq. `int4=True` composes with either.

4. Results — the headline

4.1 Pooled headline: both knob settings $\times$ {vs think, vs no-think} $\times$ {latency, FLOPs}

Reproduced **live** by the two knob settings of `method_final.py` (full suite, $n = 42\,374$, sample-weighted). `d_think` = method — always-32B-think (0.5632); `d_nt` = method — always-32B-no-think (0.5732); both baselines 4.57 FLOP-eq, latency 10 521.6 ms (think) / 665 ms (no-think). Latency seq / par = batch-1 sequential (single-stream) / parallel (2-GPU co-resident: best-of-N batched, fusion legs concurrent).

knob (V1)	vs 32B-THINK	vs 32B-no-think	FLOP-eq ($\times 32B$)	batch-1 latency seq / par (-vs think)
compute-lean (Pandora open)	+0.0117	+0.0017	2.244 (0.49 $\times$)	578 / 469 ms (-94 % / -96 %)
↳ <i>fixed-bo8 reference</i>	+0.0118	+0.0018	2.538	460 / 460 ms
accuracy-max (F3 PMC fusion)	+0.0238	+0.0137	5.695 (1.25 $\times$)	1050 / 666 ms (-90 % / -94 %)

Both modes are faster than always-32B-no-think on parallel latency and $\geq 90\%$ faster than always-32B-THINK on both accountings. The compute-lean *fixed-bo8 reference* row is the literal +0.0118; the

deployed compute-lean point swaps fixed best-of-8 for Pandora adaptive-N, which holds that accuracy iso (+0.0117) while cutting FLOPs 2.54 → 2.24.

4.2 Final per-benchmark tables (compute-lean and accuracy-max V1)

compute-lean (MCQ margin cascade + Pandora open + MMMU keep-7B):

benchmark	n	7B	32B-nt	32B-thk	policy	method	d_think	d_nt	FLOPs	lat seq / par
PMC-VQA	33 430	0.5427	0.5518	0.5494	margin-cascade	0.5508	+0.0014	−0.0010	1.386	403 / 403
SLAKE-closed	836	0.8254	0.8589	0.8636	margin-cascade	0.8516	−0.0120	−0.0072	1.935	483 / 483
VQA-RAD-closed	251	0.7809	0.8526	0.8406	margin-cascade	0.8328	−0.0079	−0.0198	3.603	726 / 726
PathVQA-closed	3 362	0.8409	0.8891	0.8891‡	margin-cascade	0.8882	−0.0009	−0.0009	3.089	651 / 651
MedXpert-MM	2 000	0.2615	0.3065	0.3040	margin-cascade	0.3005	−0.0035	−0.0060	5.095	943 / 943
MMMU-Medical	150	0.8000	0.6333	0.6600	keep-7B	0.8000	+0.1400	+0.1667	1.000	347 / 347
SLAKE-open	645	0.7364	0.8186	0.6236§	pandora-N + verifier	0.8093	+0.1857	−0.0093	7.622	1906 / 627
VQA-RAD-open	200	0.4650	0.6000	0.4800§	pandora-N + verifier	0.5950	+0.1150	−0.0050	8.391	2124 / 605
PathVQA-open	1 500	0.3240	0.3760	0.2460§	pandora-N + verifier	0.4520	+0.2060	+0.0760	12.598	3100 / 759

accuracy-max V1 (MCQ F1 slice router: PMC → F3 fusion; open cells identical):

benchmark	n	policy (F1 certified)	md_tFLOPs	lat seq / par
PMC-VQA	33 430	fusion (F3 conf-adv)	0.5508	1012 / 403
SLAKE-closed	836	always-32B-nt	0.8516	665 / 483
VQA-RAD-closed	251	always-32B-nt	0.8328	665 / 726
PathVQA-closed	3 362	always-32B-nt	0.8882	665 / 651
MedXpert-MM	2 000	always-32B-nt	0.3005	665 / 943
MMMU-Medical	150	keep-7B	0.8000	347 / 347
SLAKE / VQA-RAD / PathVQA-open	2 345	pandora-N + verifier	(as above)	

‡ PathVQA-closed has no 32B-think dump → think = no-think. § open-text 32B-think acc is **estimated** (judged 32B-no-think + measured modal think-delta −0.195 / −0.120 / −0.130).

Pooled by pool (both V1 modes):

i pool	n	method	d_think	d_nt	FLOP-eq (×32B)	lat seq (−think)	lat par (−think)	macro d_think
compute-lean	42 374	0.5749	+0.0117	+0.0017	2.244 (0.49×)	578 ms (−94 %)	469 ms (−96 %)	+0.0693
MCQ only	40 029	0.5756	+0.0011	−0.0008	1.739 (0.38×)	454 ms (−96 %)	454 ms (−96 %)	+0.0195
open only	2 345	0.5625	+0.1927	+0.0456	10.871 (2.38×)	2688 ms (−74 %)	710 ms (−93 %)	+0.1689
accuracy-max	42 374	0.5869	+0.0238	+0.0137	5.695 (1.25×)	1050 ms (−90 %)	666 ms (−94 %)	+0.0747
MCQ only	40 029	0.5883	+0.0139	+0.0119	5.392 (1.18×)	954 ms (−91 %)	664 ms (−94 %)	+0.0276
open only	2 345	0.5625	+0.1927	+0.0456	10.871 (2.38×)	2688 ms (−74 %)	710 ms (−93 %)	+0.1689

Read. The two accuracy engines are the **open-text arm** (bo8 + verifier beats even 32B-no-think, which beats 32B-think by +0.12...+0.21) and **MMMU keep-7B** (+0.140 vs think). Perception MCQ ties 32B-nt (≈ think). Per-benchmark Δ vs think: PMC +0.0014, SLAKE-cl −0.012, VQA-RAD-cl −0.008, PathVQA-cl −0.001, MedXpert −0.004,

MMMU +0.140, SLAKE-o +0.192, VQA-RAD-o +0.105, PathVQA-o +0.207.

4.3 The reconciliation: +0.0118 and +0.0238 are ONE method at two knob settings

An earlier inconsistency (the compute-lean pipeline reproduced +0.0118 while the task quoted +0.0238) is **not a contradiction** — +0.0118 is *compute-lean* (base margin router) and +0.0238 is *accuracy-max* (PMC slice-fusion). Only the MCQ arm differs; the open (Pandora) arm is byte-identical. The load-bearing cell is PMC: fusion lifts PMC acc +0.0145 (0.5508 → 0.5653), and because PMC is $n = 33\,430 = 78.9\%$ of the pooled 42\,374, that one cell adds $0.789 \times 0.0145 = +0.0114$ to the pooled sample-weighted accuracy; the four non-PMC closed cells snapping margin-cascade → always-32B add a further $\approx +0.0006$. Together pooled d_think +0.0117 → +0.0238. Macro-avg barely moves (+0.0693 → +0.0747) because PMC's +0.0145 contributes only /9 and macro is already dominated by MMMU (+0.14) and open text (+0.19). (Bonus: the unified method scores SLAKE-open with bo8 + verifier in *both* modes, so unified accuracy-max FLOPs are 5.695, cleaner than the separate-script 5.751.)

4.4 V2 — F8 + F10 make BOTH modes FLOP-negative

method_final.py::run_v2 folds two held-out levers into the *same* pipeline (V1 output unchanged): **F8** (certified weak-veto) replaces F3 in the accuracy-max PMC cell; **F10** (team-objective L2D) replaces the open arm's τ gate in both modes.

knob	vs 32B-THINK	vs 32B-no-think	FLOP-eq (×32B)	lat seq / par	FLOP-neg?
compute-lean V1 (Pandora- τ open)	+0.0117	+0.0017	2.244 (0.49×)	578 / 469 ms	[YES]
compute-lean V2 (F10 open)	+0.0123	+0.0023	2.238 (0.49×)	577 / 468 ms	[YES]
accuracy-max V1 (F3 fusion, Pandora- τ open)	+0.0238	+0.0137	5.695 (1.25×)	1050 / 666 ms	□
accuracy-max V2 (F8 veto + F10 open)	+0.0212	+0.0112	4.246 (0.93×)	839 / 729 ms	[YES]

Both modes are now FLOP-negative (< always-32B's 4.57): compute-lean **0.49×** (always was), accuracy-max **newly 0.93×** (was 1.25×). The accuracy-max FLOP cut is **−1.449 FLOP-eq (−25 %)** for a **−0.0026** vs-think give-back (F8 captures 70 % of F3's PMC beat; F10 gives some back on open cells) — a strict Pareto move into the compute-negative half-plane while still CI-certified above 32B on PMC. Honest trade: accuracy-max parallel latency rises 666 → 729 ms because F8 is a *sequential* cascade on PMC (7B → maybe 32B, no parallel-fusion overlap) — still **−93 %** vs 32B-think. **accuracy-max V2 is all-axes dominant over the naive baseline: cheaper (0.93×), faster (−93 %), and more accurate (+0.0212).**

4.5 The latency headline

Across settings the method beats always-32B-THINK on batch-1 latency by **−90 % to −96 %** (parallel), and it is *faster than the cheap no-think 32B baseline* on parallel latency in both modes — the G8 prefetch and the parallel best-of-N leg together keep every escalation-heavy cell under the 665 ms no-think anchor. The sequential (single-stream) accounting is more conservative (adaptive draws are one-at-a-time, fusion legs serial) but still **−74 % to −96 %**.

4.6 The ACC reframe vs always-32B-THINK — five families (efficiency engine)

With the think baseline priced (\$2.4), the ACC MCQ bake-off (master_data.csv, measured batch-1) is re-scored against **always-32B-THINK** (reframe_vs_bigthink.json). **Always-32B-THINK is Pareto-dominated on every family:**

f pool	32B-think acc	method acc	Δ acc	32B-think lat	method lat	Δ lat	FLOPs%	32B-think E	method E	think-esc%
MVT										
ALL-6 7B→32B	0.5723	0.5693	−0.003	11.34 s	2.27 s	−80 %	52 %	6319 J	1182 J	15 %
ALL-5	0.6463	0.6450	−0.001	8.88 s	0.44 s	−95 %	25 %	4916 J	173 J	2 %
Lingshu										
ALL-6 7B→32B	0.6611	0.6614	+0.000 (beats)	0.32 s	0.30 s	−8 %	49 %	113 J	76 J	0 %
ALL-5	0.7746	0.7726	−0.002	0.32 s	0.25 s	−24 %	39 %	113 J	61 J	1 %
QiuQ-Med										
ALL-6	0.4689	0.5095	+0.041 (beats)	9.72 s	0.12 s	−99 %	9 %	5382 J	18 J	0 %
ALL-5	0.5432	0.6048	+0.062 (beats)	8.49 s	0.12 s	−99 %	9 %	4692 J	18 J	0 %
Chiton										
ALL-6	0.5076	0.6023	+0.095 (beats)	4.25 s	0.20 s	−95 %	19 %	1176 J	17 J	0 %
ALL-5	0.5926	0.7249	+0.132 (beats)	3.66 s	0.20 s	−95 %	20 %	1006 J	17 J	0 %
MedGemma										
ALL-6	0.5253	0.5219	−0.003	12.72 s	3.37 s	−74 %	68 %	6535 J	1614 J	20 %

ALL-5	0.5979	0.6028	+0.005 (beats)	9.76 s	0.18 s	−98 %	12 %	4990 J	16 J	0 %
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On **3 of 5 families the big *thinking* model is actually LESS accurate than the method** (QoQ +0.041, Chiron +0.095) — think over-thinks the perception-dominated suite — *and* 1.2–45× slower. think : no-think latency ratio per family (ALL-6): MVT 49×, MedGemma 45×, QoQ 43×, Chiron 15×, Lingshu 1.2× (Lingshu has no real promptable think mode, hence its small latency win).

4.7 The trained-verifier accuracy engine — three families beat the 32B, plus boxes

The open-text arm's verifier is a positive result in its own right. Across **three families and two architectures**, the 7B-verifier cascade **beats the strong 32B/38B** on accuracy on every family × dataset cell (pooled cascade-best vs STRONG): **Lingshu 0.421 vs 0.331, MedVLThinker 0.344 vs 0.277, InternVL3 0.255 vs 0.218** — the InternVL3 win uses the *Lingshu-trained* verifier applied cross-architecture (no retraining). A strict same-split test (pool the four Lingshu open sets, hold out 30 %, n = 1 064) orders **verifier 0.501 > 32B 0.462 > SC 0.411 ≈ greedy 0.413** — the trained 7B captures **49 %** of the oracle gap and edges the 32B *while 5× smaller*. Two-seed honesty: the verifier beats the same-split 32B by **+0.039, 95 % CI [+0.010, +0.066]** on seed-0 but ties on seed-1 (−0.005), so the honest verb versus the 32B is **matches-to-modest-win (mean +0.017)**; versus *training-free* selection it is robustly positive both seeds (+0.066 / +0.088). The verifier separates correct from incorrect candidates at **AUROC 0.924**, a blank-image ablation drops it −0.047 (it reads the image), and a trained 7B verifier beats a *zero-shot* 32B verifier (0.403 vs 0.355) — **task training beats 5× scale for selection**. The same recipe transfers to **bounding boxes**: SLAKE organs recover **40 % / 53 %** and the real MS-CXR chest-X-ray benchmark **78 % / 77 %** of the oracle gap (MS-CXR bootstrap gain **+0.191, 95 % CI [+0.152, +0.232]**, a 5.6× lift).

The three-family beats-32B master tables. Each family is evaluated on its full open-ended sets; "cheap(SC)" is the small model's self-consistency majority, "STRONG" the family's single 32B/38B pass, "verifier-bo8" best-of-8 selection with no gate, "cascade-best@esc" verifier-bo8 + the verifier-confidence gate escalating the stated fraction, "oracle@8" the luck ceiling. The verifier base is **Lingshu-7B** throughout; the InternVL3 column applies the *same Lingshu-trained verifier* to InternVL3 answers (cross-architecture transfer).

Lingshu (strong = Lingshu-32B)	cheap(SC)	STRONG	verifier-bo8	cascade-best @esc	oracle@8
VQA-RAD	0.465	0.600	0.575	0.625 @24 %	0.630
PathVQA	0.324	0.376	0.453	0.469 @33 %	0.517
Kvasir	0.286	0.301	0.439	0.448 @20 %	0.491
RadImageNet-OOD	0.329	0.289	0.353	0.353 @0 %	0.512
POOLED	0.322	0.331	0.414	0.421 @12 %	0.513

MedVLThinker (strong = MVT-32B)	cheap(SC)	STRONG	verifier-bo8	cascade-best @esc	oracle@8
VQA-RAD	0.420	0.525	0.490	0.555 @54 %	0.600
Kvasir	0.343	0.361	0.477	0.483 @9 %	0.550
RadImageNet-OOD	0.204	0.202	0.241	0.243 @13 %	0.317
POOLED	0.266	0.277	0.339	0.344 @14 %	0.416

InternVL3 (strong = IV3-38B; Lingshu-trained verifier)	cheap(SC)	STRONG	verifier-bo8	cascade-best @esc	oracle@8
VQA-RAD	0.445	0.415	0.570	0.580 @12 %	0.620
PathVQA	0.081	0.096	0.116	0.125 @52 %	0.192
Kvasir	0.362	0.380	0.479	0.487 @17 %	0.593
RadImageNet-OOD	0.285	0.304	0.302	0.313 @52 %	0.398
POOLED	0.202	0.218	0.249	0.255 @36 %	0.337

Two honest qualifications. **(i) The gate does real work where the strong model is genuinely better.** Pure verifier-bo8 (no gate) does not uniformly beat STRONG (it loses Lingshu VQA-RAD 0.575 < 0.600, MVT VQA-RAD 0.490 < 0.525, IV3 RadImageNet 0.302 < 0.304); on exactly those sets the gate escalates a larger fraction (24 %, 54 %, 52 %) and the cascade climbs above STRONG by combining best-of-N on recoverable questions with deference on the rest — it wins by *knowing when to escalate*, not by out-selecting the 32B everywhere. **(ii)** The pooled STRONG numbers are low because these full pools are dominated by hard/OOD sets and exclude SLAKE (the one set where the 32B dominates); the stricter same-split test below puts the 32B back at 0.462.

The canonical same-split test (n = 1 064). Pool the four Lingshu open sets, hold out 30 %, evaluate every method — including the 32B — on the *identical* held-out questions:

dataset	greedy	SC	32B (same split)	verifier	oracle	n
SLAKE	0.738	0.738	0.829	0.762	0.895	210

VQA-RAD	0.519	0.500	0.648	0.611	0.722	54
PathVQA	0.352	0.349	0.377	0.441	0.513	435
Kvasir	0.282	0.282	0.326	0.405	0.493	365
POOLED	0.413	0.411	0.462	0.501	0.592	1 064

Ordering **verifier 0.501 > 32B 0.462 > SC 0.411 $\approx$ greedy 0.413**: the verifier captures **49 %** of the oracle gap and edges the 32B *while 5 $\times$ smaller*, beating it on the hard sets (PathVQA 0.441 vs 0.377, Kvasir 0.405 vs 0.326) and losing where the 32B is genuinely stronger (SLAKE 0.762 vs 0.829) or *n* is tiny (VQA-RAD, *n* = 54). **Two-seed honesty**: seed-0 paired-bootstrap gain over the 32B **+0.0385, 95 % CI [+0.010, +0.066]** (significant); seed-1 **-0.005** (a tie). Honest verb: **matches-to-modest-win** (mean +0.017); robust only over *training-free* selection (+0.088 / +0.066 over greedy both seeds; gap-captured 35–49 %, mean $\sim$ 42 %, wide per-dataset spread: SLAKE 15.2 %, VQA-RAD 45.5 %, PathVQA 55.7 %, Kvasir 58.4 %).

Training is the active ingredient; argmax is the rule. Zero-shot self-verification is luck-floored (AUROC $\approx$ 0.5); training the same 7B on judge labels lifts selection to 0.501, and the **trained 7B verifier beats a zero-shot 32B verifier 0.403 vs 0.355** (selection efficiency 0.810 vs 0.717) — task training beats 5 $\times$ the parameters. The verifier separates correct from incorrect at **AUROC 0.924** (mean score 0.749 vs 0.171 across 8 512 candidates); a blank-image ablation drops discrimination **-0.047** (it reads the image, refuting the "lazy verifier" failure mode). Plain argmax (0.501) beats verifier-weighted voting (0.489) and a score $\times$ count hybrid (0.470) — any vote-flavored rule reinherits the majority trap. Best-of-K scales monotonically (*K* = 1/2/4/8 $\rightarrow$ 0.385/0.425/0.476/0.501, oracle@8 0.592) while a random pick stays flat — a genuine **test-time-scaling** property; and because $2N < 4.57$ for $N \leq 2$, **verifier-bo2 (no gate) beats the 32B on accuracy and FLOPs (4.0 vs 4.57)** on the hard/OOD sets. **Structured outputs (boxes)**: with *y* = 1[IoU $\geq$ 0.3] and a Qwen2.5-VL-7B box verifier, best-of-8 recovers **40 % / 53 %** (SLAKE organs, two seeds; both training-free selectors *below* greedy) and **78 % / 77 %** (MS-CXR, two seeds; +0.191 over greedy, 5.6 $\times$ lift) of the oracle gap — training again roughly doubles the zero-shot box-verifier's captured gap (0.115 $\rightarrow$ 0.232). **The binding limit is candidate quality**: per-answer AUROC is already $\sim$ 0.90, and a Bradley-Terry ranking loss pushes it to 0.93, yet selection stays flat at $\sim$ 0.50; what moves the ceiling is better candidates (cross-model pooling lifts oracle@8 +0.11–0.15).

5. Ablations — where each piece earns its place

5.1 Gate bake-off — margin > agreement > CASP on Lingshu (and it is family-dependent)

Repeated from §3.3 as the load-bearing ablation. On Lingshu MCQ (*n* = 37 879) the deployed **margin** gate wins on the deployable metric (min-esc 15.62 % to 32B-nt parity, AUROC 0.7254); **CASP-stability is inert** (7B is 98.95 % cap320-vs-full stable), **agreement** is the worst ranker and needs the 32B, **conf/MSP** ties AUROC but needs 20.3 % escalation. **Family-dependence (the honest nuance)**: on **MedVLThinker** — where the 7B is resolution-sensitive — a *trained* CASP-Stability gate beat the confidence gate at ALL-6 parity (FLOPs **49.0 % vs 53.9 %** margin / 57.4 % MSP, latency 1.77 s vs 2.69 s, acc 0.5698 vs 0.5687), because re-targeting the routing label from un-learnable recoverability (AUROC 0.58, the wall) to learnable **stability** (AUROC 0.71) breaks the training-free ceiling (capacity is irrelevant: logistic $\approx$ MLP $\approx$ LoRA). So "no trained gate beats confidence" holds only for gates predicting *recoverability*; a trained *stability*-router beats confidence on cascade efficiency where the cheap model is unstable. The MedEvalKit gate bake-off (gate_unified_bakeoff.json) makes the deeper point: **cascade quality (ADC) tracks recoverability-AUROC, not detection-AUROC** — on MCQ competent-4 the highest-*detection* gate (learned-RICH 0.693) has *lower* ADC than margin, and the only positive-routing-eff gate is 7B self-verify (AutoMix, the highest *recoverability* AUROC 0.614).

5.2 The middle-tier ablation (M1/M2/M3) — it is the structure, not the gate

The load-bearing structural fact (METHOD_ACC.md head-to-head, measured latency/energy, MVT ALL-6): inserting the **32B-no-think middle tier** turns an escalate-everything-to-think cascade into the ACC.

variant (ALL-6)	acc	think-esc	FLOPs%	latency	energy
M2 escalate-to-think, no nt-middle (7B-think $\rightarrow$ 32B-think)	0.5725	86 %	105 %	29.8 s	7 049 J
M3 7B-think middle (7B-nt $\rightarrow$ 7B-think $\rightarrow$ 32B-think)	0.5697	65 %	89 %	23.2 s	5 499 J
M1 ACC — 32B-nt middle restored	0.5694	19 %	55 %	5.9 s	1 505 J
M1b ACC + agreement gate	0.5710	14 %	54 %	4.86 s	1 220 J

Δ (M1 vs M2): think-esc 86 % $\rightarrow$ 19 %, FLOPs 105 % $\rightarrow$ 55 %, latency **29.8 $\rightarrow$ 5.9 s (-80 %)**, energy -79 %, at matched accuracy. Complementarily, holding the three-tier structure fixed and **swapping in every training-free gate barely moves the operating point** (ALL-6 band: acc 0.5666–0.5702, FLOPs 49–62 %, latency 1.77–3.48 s); only *random*

escalation leaves the frontier (116.5 % FLOPs). The gate moves you a few points along a frontier the structure already fixed — the empirical face of the recoverability wall.

The three-tier cost table (why avoiding a think call dominates). Per-tier measured batch-1 costs (one forward = $2 \cdot N \cdot (P+G)$, $N_7 = 7.6e9$, $N_{32} = 33.0e9$):

tier	config	prefill P	decode G	FLOPs (×1e15)	latency	energy
T0	7B no-think @ cap320	388	2	0.01	0.21 s	25 J
T1	32B no-think @ cap320	388	2	0.03	0.34 s	65 J
T2	32B think @ fullres	685	391	0.07	11.34 s	~6 319 J

The two no-think tiers are *constant-energy* (~25 J, ~65 J, ~2 decode tokens); the think tier's energy scales with its decode length ($E \approx 18.17 \cdot G - 107.5$ J), so the whole cascade cost is governed by $e_1 \cdot c_{T2}$ — *how often T2 fires*. A conventional 2-tier 7B → 32B-think cascade at parity sends $e_1 \approx 69$ % to think; ACC's free no-think middle tier cuts that to **15 %**. The over-thinking premise that justifies the middle tier (MedVLThinker 32B, exact-match): **SLAKE 32B-nt 0.849 vs 32B-think 0.764 (+0.085)**, VQA-RAD **0.853 vs 0.776 (+0.077)**, PMC/PathVQA flat; pooled competent-4 **32B-nt 0.658 ≥ 32B-think 0.645** at ~2 vs ~477 decode tokens. Think *fixes* 11.5 % of no-think's errors but *breaks* 11.1 % of its correct answers — symmetric over-thinking, not a free lunch. And no-think@cap320 recovers think@fullres accuracy (0.6463 vs 0.6451) at **39 %** of the compute, so cap320 is the natural T1.

ALL-6 all-methods bake-off at parity (MedVLThinker) — the full frontier (every training-free gate in the same 3-tier structure; guard = # benchmarks worse than always-7B, averaged over 20 held-out seeds):

system	acc	esc ₀	think	FLOPs%	latency	energy	guard
always-7B-nt @cap320	0.5262	0 %	0 %	8.4 %	0.13 s	19.9 J	0.00
always-32B-nt @cap320	0.5573	100 %	0 %	36.2 %	0.23 s	77.8 J	0.00
always-32B-think [PARITY]	0.5723	100 %	100 %	100.0 %	11.34 s	6 318.8 J	0.00
ACC-v2 (agreement gate)	0.5693	71.7 %	15.1 %	52.0 %	2.27 s	1 181.9 J	0.00
ACC-v1 (margin gate)	0.5687	66 %	19 %	53.9 %	2.69 s	1 416.6 J	0.00
CASP-Stability (trained gate)	0.5698	74 %	11 %	49.0 %	1.77 s	899.2 J	0.05
MSP/Chow	0.5697	70 %	19 %	57.4 %	2.96 s	1 568.1 J	0.00
entropy	0.5691	71 %	21 %	62.0 %	3.48 s	1 863.1 J	0.00
Gini/DOCTOR	0.5702	69 %	22 %	61.0 %	3.44 s	1 837.1 J	0.00
AutoMix (self-verify)	0.5692	73 %	18 %	54.6 %	2.50 s	1 307.0 J	0.05
FrugalGPT-style learned	0.5677	70 %	19 %	60.4 %	3.30 s	1 765.5 J	0.10
Jitkrittum L2D (Diff-Prob)	0.5666	67 %	15 %	50.6 %	2.29 s	1 194.5 J	0.00
random	0.5641	89 %	76 %	116.5 %	8.95 s	4 889.5 J	0.05

Every training-free gate lands in a narrow band (acc 0.5666–0.5702, FLOPs 49–62 %, latency 1.77–3.48 s); only random leaves the frontier. **Per-benchmark, ACC-v2 is ≥ the 7B on all seven columns (guard 0)** and beats the think baseline on the four perception sets (PMC 0.561 vs 0.556, SLAKE 0.842 vs 0.764, VQA-RAD 0.861 vs 0.776, PathVQA 0.679 vs 0.673), trailing only on the three reasoning sets (MMMU 0.643 vs 0.688, MedXpert-R 0.282 vs 0.326, MedXpert-U 0.310 vs 0.384) — the honest scope boundary. On **ALL-5** (drop MedXpert) always-think **8.88 s / 4 915.9 J** → **ACC-v2 0.44 s / 172.8 J, FLOPs 24.9 %** (–95 % latency, ~28× energy, guard 0.05).

Five-family generalization (no-think ≥ think on perception re-measured across three architectures):

family	set	parity acc (32B-think)	ACC-v2 acc	ACC-v2 FLOPs%	esc ₀	think	guard
MedVLThinker	ALL-6 / ALL-5	0.5723 / 0.6463	0.5693 / 0.6450	52.0 / 24.9	71.7 / 35.1	15.1 / 2.3	0.00 / 0.05
Lingshu	ALL-6 / ALL-5	0.6611 / 0.7746	0.6614 / 0.7726	48.6 / 38.9	61 / 42	0 / 1	1.00 / 1.15 (inherited)
QoQ-Med	ALL-6 / ALL-5	0.4689 / 0.5432	0.5095 / 0.6048	8.8 / 9.1	0 / 0	0 / 0	0.00
Chiron-o1	ALL-6 / ALL-5	0.5076 / 0.5926	0.6023 / 0.7249	19.3 / 19.9	0 / 0	0 / 0	0.00
MedGemma	ALL-6 / ALL-5	0.5253 / 0.5979	0.5219 / 0.6028	68.4 / 11.5	55 / 0	20 / 0	1.15 / 0.00

Lingshu is a clean win (roughly halving FLOPs; the guard 1.00 is *inherited* from the target — always-32B is itself worse than Lingshu-7B on MMMU, the anomaly, so any system reaching for the 32B pays that guard). **QoQ / Chiron gracefully collapse** — think *hurts* there, so the agreement gate never escalates (esc 0 %), delivering accuracy *above* the think target at cheap-leg cost. **MedGemma is the honest partial case** (ALL-6 over-escalates for 68.4 % FLOPs; ALL-5 collapses to 11.5 %). ACC pays where the 32B has a genuine think-mode advantage that over-fires on perception; elsewhere it degrades gracefully.

5.2b The faithful cross-family MCQ margin cascade (−17...−69 % FLOPs)

The simplest instance of the ACC structure — a **2-tier** cheap-7B → strong-32B margin cascade at the iso-strong operating point ($c = 1 + \text{esc} \cdot 4.57$) — reproduced under the faithful MedEvalKit protocol matches the strong model at large FLOPs savings, with the win magnitude tracking the (strong − cheap) accuracy gap:

benchmark	Lingshu	MedVLThinker	InternVL3
MMMU-Med	keep-7B (anomaly)	−14 %	−62 %
PMC-VQA (33k)	−69 %	−49 %	−16 %
SLAKE	−56 %	no win (7B weak)	keep-cheap (8B ≥ 38B)
VQA-RAD	−17 %	−41 %	−67 %
PathVQA	−31 %	−68 %	−20 %
MedXpert-MM	no win (floor)	no win	(cap gap)

A *small* gap → thin escalation residual → large saving; a *large* gap (weak cheap model) → escalate nearly everything → *more* expensive than always-strong. Two principled non-winning regimes: **keep-cheap** (IV3-SLAKE 8B ≥ 38B; Lingshu-MMMU inflated 7B) and **no win** (MVT-SLAKE 7B 0.498 vs 32B 0.620 forces 96 % escalation, +18 % FLOPs; every MedXpert cell near the floor, ~100 % escalation). **Threshold honesty**: the −69 % PMC-Lingshu figure is at the iso-strong operating point; under a *fair held-out* τ it is **−57 %** (cascade 0.563 ≥ 32B 0.549), and under oracle- τ −74 % — the deployable, honest figure is the middle one. **The mirror-image lever, a reasoning think tier**, adds accuracy where headroom exists (MMMU-Med Δ **+0.034 Lingshu** / **+0.107 MVT** / **+0.120 IV3**; MVT-MedXpert +0.045; Lingshu-MedXpert ~0 at the floor), gated so MMMU reaches think accuracy at ~78 % FLOPs (a win) while MedXpert lands at 143–151 % (no win) — regime-adaptive.

5.3 F3 vs F8 on the accuracy-max PMC cell

policy	PMC acc	d vs 32B-nt	veto %	esc → 32B %	PMC FLOP-eq	lat seq / par	note
F3 confidence-advantage fusion (V1)	0.5653	+0.0135	—	100 %	5.570	1012 / 665 ms	both legs on 100 % of PMC
F8 certified weak-veto (V2)	0.5613	+0.0095	40.0 %	60.0 %	3.741	746 / 746 ms	7B on all + 32B on non-veto

F8 captures **70.4 %** of F3's PMC beat at **−32.8 % PMC FLOPs**, still CI-certified above 32B ([0.0071, 0.0118]). Classic per-slice-reliability Chair-Varshney collapses to always-32B ($d = 0.0$); the beat requires per-sample confidence.

5.4 F10 open-text repair (team objective vs parity- τ)

open cell	n	prior gate (Pandora- τ)	F10 L2D	gain	repaired?	F10 acc	esc %	FLOP-eq
SLAKE-open	645	−0.0093	+0.0016	+0.0109	[YES]	0.8202	20.6 %	7.842
VQA-RAD-open	200	−0.0050	+0.0050	+0.0100	[YES]	0.6050	37.0 %	9.511
PathVQA-open	1 500	+0.0760	+0.0860	+0.0100	(already won)	0.4620	26.5 %	12.178

All three lift; the two residual open losses are repaired from below-32B to above-32B. Pooled open-only: acc 0.5625 → 0.5727, d vs 32B-nt +0.0456 → +0.0559, d vs think +0.1927 → +0.2029, at slightly lower FLOPs (10.871 → 10.758). The gain is from the objective, not a better signal (the CI on the small VQA-RAD-open / SLAKE-open point-beats still spans 0; the robust open beat remains PathVQA-open).

5.5 Pandora and its refinements

At **iso-bo8 accuracy (held out)** Pandora reaches the target at **−27 % FLOPs** / **−28 % energy** vs fixed best-of-8, beating even the *optimistically-tuned* adaptive-N (−19 %) and gate (0 %) baselines, despite being the only method whose thresholds are held-out:

Method	ds covered	FLOPs	vs bo8	energy (J)	meanN	esc	lat_seq (ms)
Pandora iso-bo8 (held-out)	9/11	11.74	−27 %	409.8 (−28 %)	5.38	21.6 %	2 951
adaptive-N (oracle- τ)	11/11	13.01	−19 %	459.5	6.31	8.7 %	3 350

verifier-bo8 + gate (oracle- τ)	11/11	16.00	0 %	568.8	8.00	0.0 %	4 176
Pandora iso-strong (held-out)	11/11	6.32	-61 %	222.6	3.03	5.5 %	1 619

The reframe Pandora forces. On these OOD open-text sets the accuracy ceiling worth matching is the **cheap best-of-8 ensemble, not the 32B** (pooled Lingshu bo8 0.414 vs 32B 0.331) — so escalating to the 32B buys little and the lever is reaching bo8 accuracy more cheaply. **Refinements (regime-limited):** correlated-Pandora (diversity-discounted stopping) is a **free -16-17 %** in meanN/FLOPs/latency/energy at iso-strong (11/11) but does *not* help at iso-bo8 (coverage drops to 7/11); Pandora $\times$ cross-model pooling lifts oracle **+0.11-0.15** but yields no frontier gain (the selection wall blocks the conversion). Honest latency trade: adaptive drawing is sequential, so Pandora's $\text{lat}_{\text{seq}} \approx 2\,951$ ms vs a batched fixed bo8's $\text{lat}_{\text{bat}} \approx 522$ ms — Pandora wins FLOPs + energy, loses batch-1 wall-clock to a batched bo8.

5.6 Escalation levers (G8, G5/G6, G3/G4/G7/G2)

Covered in §3.5. G8 slice-gated + G5(ϵ^*) gives pooled **416.4 ms (-96.0 % vs 32B-think)** at -0.0018 acc with no cell slower than always-32B-nt; G3 INT4 is a projected VRAM/energy lever (FLOPs unchanged under MAC accounting); G4/G7/G2 are data-limited on the dumps and not deployed.

5.7 Router vs unified verdict

A **deterministic format-aware router** (`unified_router.json`, dispatch MCQ $\rightarrow$ margin cascade, open $\rightarrow$ verifier best-of-8 cascade, score the *whole pooled* stream as one point) is a **tested Pareto win over always-strong on all three families**: Δacc **+0.003 (Lingshu) / +0.000 (MVT) / +0.001 (IV3)** at **48-82 % of always-32B FLOPs** (30-59 % if the image prefill is prefix-shared across the 8 samples). The FLOPs saving tracks how competitive the strong model is (Lingshu's 32B barely ahead $\rightarrow$ keep-cheap $\rightarrow$ biggest saving; IV3's 38B more competitive $\rightarrow$ more escalation $\rightarrow$ smallest). A single *unified* gate is beaten by margin on MCQ and verifier-conf on open $\rightarrow$ keep the two-arm router (Correction #2).

5.8 The reframe measurements (the two blocked cells, now measured)

The whole reframe rested on the cost of the strong leg. Both previously-blocked cells were measured (§2.4): **open-text 32B-THINK = 10 521.6 ms / 2 001.9 J** (15.8 $\times$ the no-think leg, and -0.15 less accurate on perception — "open-text think 10.5 s / -0.15"), and **IV3-38B = 6 220 ms / 3 275.6 J think** ("IV3 6.2 s", removing the router's amortized-batch-latency proxy caveat for the 38B peer). Think-hurts-perception is consistent across families (SLAKE: MVT -0.084, Chiron -0.108; VQA-RAD: all -0.07...-0.09). These turn the ledger's "blocked" cells into measured constants; OmniMed-32B stays blocked, so ALL-7 is unreportable and ALL-6 is the computable pool.

5.9 Best-of-N internals — UGV, diverse generation, pairwise verifier, portfolio/pooling, ceiling-break

- **UGV (single generative verifier) for MCQ — negative.** Scoring MCQ options as *generated answers* collapses accuracy: PMC-VQA strict greedy **0.132 content vs 0.534 letter** (-0.394); content-MCQ verifier gain +0.004, AUROC ≈ 0.70 . The router stays the better method (MCQ $\rightarrow$ letter + margin; open $\rightarrow$ trained bo-N verifier); the generative verifier's home is open-text. Backlog B2 closed as a negative for MCQ.
- **Diverse generation (GPU) — positive on coverage.** A 5-persona $\times$ 3-temperature portfolio ($M = 15$) lifts the oracle ceiling **+0.027** at matched budget / **+0.064** at $M = 15$ (CIs exclude 0), converting to a significant **+0.025 verifier bo-N accuracy** at 1.875 $\times$ generation cost. It shifts the binding limit from coverage (#1) to selection (#2): on PMC-content the oracle lift is largest (+0.110) but the pointwise verifier cannot convert it (confident-distractor rate 0.426 $\rightarrow$ 0.504).
- **Real pairwise verifier (GPU) — positive on selection; overturns the simulation.** A real A-vs-B verifier beats pointwise-argmax on selection (sel_acc 0.374 $\rightarrow$ **0.410, +0.036 CI** [0.016, 0.055]; on near-ties +0.050), closing ~ 35 % of the pointwise $\rightarrow$ oracle gap; knockout captures ~ 87 % of the win at ~ 7 comparisons/q. This overturns a *simulated*-pairwise parity (deriving $P(i>j)$ from pointwise scores cannot manufacture comparative signal; a real forward pass carries it).
- **Cross-model pooling / portfolio.** Pooling 3 cheap generators lifts oracle **+0.08 (B = 8) / +0.11 (B = 16)** held-out, but the Markowitz-optimal allocation $\approx$ a naive uniform split (Δ +0.002 to +0.02, negative on `vqa_rad`) — **the win is diversity/pooling, not the clever allocation** (the bandit test reaches the same verdict independently). Error- $\phi \approx 0.52$ -0.56 (models fail on somewhat different questions, still positively correlated).
- **Diversity selection (DPP/MMR) — a rate win, not a ceiling lift.** Reordering a fixed 8-candidate set leaves oracle@8 identical (0.412) by construction, but MMR reaches the first correct answer with -15.6 % fewer samples; only 4.67 of 8 candidates are distinct (≈ 42 % exact duplicates).
- **The open-ended ceiling-break.** The pessimistic ~ 0.6 gate AUROC is partly an MCQ discreteness artifact: the *same* Lingshu-7B margin signal rises to **AUROC ≈ 0.87** for detecting cheap errors on free-text (pooled 0.846, 95 % CI [0.830, 0.862]; SLAKE 0.889, PathVQA 0.797, VQA-RAD 0.717) — and it is discreteness, not length (token-F1 agrees).

But **recoverability does not break** (pooled 0.591), and even open-ended, **plain confidence stays the best training-free gate** (fusion of all signals ties it at 0.866; self-verify P(True) 0.755 is worse). This is exactly why a *trained* verifier is necessary, and why medical-VLM cascades should be evaluated open-ended.

6. The two walls — the honest headline negative

The program is bounded by two limits. We present them as a *headline contribution*, not a footnote, because knowing exactly where the method stops is what makes the deployable claims trustworthy — and because the attempts to break each wall are themselves informative. The walls are decisions about *which model answers*; the method always produces an answer.

6.1 The recoverability wall — the broad-slice MCQ beat is confined to PMC (six independent confirmations)

A cascade gate needs **recoverability** — whether the *strong* model will fix a specific cheap error — not mere cheap-model-wrongness. Over a frozen pair, recoverability is nearly unlearnable: across **12 signal families** (margin, MSP/Chow, entropy, Gini/DOCTOR, energy, a hidden-state probe, self-verification P(True), conformal/CP-Router, a learned GBM, cross-model agreement, multi-resolution stability, semantic self-consistency) recoverability AUROC sits at **0.5-0.69**. The cause is **error correlation** — the two models fail together. On the competent-4, decomposing every question by the joint (7B, 32B) outcome: beneficial (7B wrong, 32B right) 15.7 %, futile (both wrong) 22.0 %, harmful (7B right, 32B wrong) 13.4 %; **recoverability P(32B right | 7B wrong) = 41.6 %**, i.e. **P(32B wrong | 7B wrong) = 0.584** (ALL-6 $\phi = 0.372$). This is exactly the regime Jitkrittum et al. (NeurIPS'23) predict a wall.

The *positive* corollary — "can we **beat** the 32B on a broad slice by fusing the two legs?" — is bounded by the same wall. **The beat exists on PMC-VQA and nowhere else broad**. We attacked this with **six independent fusion/routing methods**, and all six certify the same two cells (PMC fusion + MMMU keep-7B) and **no genuinely-new one**:

#	method	source	PMC d vs 32B-nt (CI)	MMMU	any NEW certified cell?
1	F3 confidence-advantage fusion (= 2-detector Chair-Varshney)	beat32b_fusion.json	+0.0135 [0.0100, 0.0169]	+0.167	no
2	F8 certified weak-veto (Wilson-LB, one-sided)	beat32b_more.json	+0.0095 [0.0071, 0.0118]	(keep-7B)	no
3	F11 Bayesian model averaging (EM additive mixture / PoE)	beat32b_more.json	+0.0134 [0.0102, 0.0165]	+0.193	no
4	F6 contrastive decoding (option-logit level)	logit_fusion.json	$\approx +0.000$ (F6_cd d = 0.0)	—	no
5	option-logit fusion (λ, α grid, F11_fixed/rw)	logit_fusion.json	+0.008 [-0.020, 0.036] (not certified)	—	no
6	Domino / H4 automatic slice discovery (106 slices x 8 splits)	robust_slice_routing.json	(re-finds PMC)	(re-finds MMMU)	no

Reading (1)–(3): F3, F8, and F11-BMA all beat the 32B on PMC (comparably-skilled, de-correlated errors) and all **gate off** on SLAKE/VQA-RAD/PathVQA-closed/MedXpert (where the 32B is clearly better — fusion *hurts* there, so the guardrail keeps 32B). F7 super-learners agree (logistic +0.0127, GBM +0.0158 on PMC; ≤ 0 elsewhere). Reading (4)–(5): fusing at the **option-logit** level (contrastive decoding, weighted logit sums) does not certify a new beat beyond PMC either — on a re-dumped per-option subsample PMC F6_cd d = 0.0 and F11_fixed +0.008 is not certified (CI spans 0). Reading (6), the decisive one: an **FDR-controlled automatic slice-discovery** search over 106 candidate slices (dataset id + question topics + 7B confidence) across 8 independent 50/50 discover/confirm splits **re-finds the PMC-fusion and MMMU-keep-7B wins without being told they are special** (validating the hand-gate) but finds **essentially no genuinely-new beat-32B slice**: genuinely-new BH-FDR5 certifications average **0.25 per split** (2 total across 8 splits, none recurring in a majority), *below* the label-permutation-null floor (mean 5.61 spurious certs, p95 = 15). **A genuinely-new slice inside an always-32B dataset does not exist above noise — the sixth confirmation of the recoverability wall**. (A real BiomedCLIP image+text embedding would sharpen the slice geometry and is the clear next upgrade; the current feature space is dataset id + question text + 7B confidence.)

MMMU is a route-to-7B anomaly, not a fusion win. The MMMU cell certifies as **always-7B** (d = +0.167, CI [0.087, 0.247], n = 150) because Lingshu-7B is inflated there (\$2.3), not because a clever fusion of two comparably-skilled legs pays off. Fusing the legs on MMMU would give the 7B ~91 % weight (F11 mean_weight_7b = 0.912) — i.e. it degenerates to keep-7B. We therefore never claim MMMU as a broad-slice fusion beat; the genuine broad win is **PMC**

fusion, and the honest ceiling is that it does not extend.

6.2 H8 credibility shrinkage validates the deployed CI-guardrail

Could a fancier actuarial estimator squeeze more certified slices out of thin data? No — and the audit is useful because it confirms the deployed rule is the right one. **Naive point routing** (deviate iff the raw discovery advantage > 0) overfits thin slices: on a fine 61-slice family it yields **~7.5 held-out guardrail violations per split**. **Hierarchical Bühlmann-Straub credibility shrinkage** (shrink each slice's advantage toward its parent-dataset mean, which is ≤ 0 for the always-32B datasets) helps only **marginally (7.5 → 6.6)** — an MSE-optimal shrinkage constant ($\bar{k} \approx 244$) is too weak to flip a raw +0.05 thin-slice noise toward a mildly negative parent, and shrinking whole-dataset cells toward the PMC-contaminated positive grand mean can even slightly worsen them. **The decisive robustifier is the simple CI lower-bound guardrail** (deviate iff the discovery 95 % lower-CI > 0 — exactly F1's existing rule): it drives fine-family violations to **0.2** and hand-family to **0.0** at a preserved pooled beat (+0.0117). **Verdict:** credibility shrinkage does not beat the CI-guardrail the method already deploys; H8's value is diagnostic — it confirms the thin-slice-overfit risk is real *and* that the CI-guardrail (not a fancier estimator) is the correct, sufficient, guardrail-honest fix.

6.3 The selectability wall — best-of-N's oracle → selection gap is largely fundamental

The open-text accuracy engine (best-of-N + trained verifier) is bounded by a *selection* ceiling. Its clearest, most surprising face is **training-free** selection: sample one open model N = 8 times and a correct answer is frequently *present* (SLAKE-open greedy 0.730 → oracle@8 0.879, headroom +0.149), yet no frozen-model rule can say *which* of the eight it is — every training-free selector ties or trails a random pick (0.720):

selector (SLAKE-open, Lingshu-7B, n = 645, MVT-32B judge)	acc	vs random	% gap-above-random captured
random pick (mean sample accuracy)	0.720	—	—
self-verify P(Yes) argmax (7B)	0.715	−0.005	worse than random
self-consistency majority	0.736	+0.016	10 %
32B pointwise verify, argmax	0.746	+0.026	16 %
32B listwise select	0.758	+0.038	24 % (best training-free)
learned fusion [self, 32B] (5-fold CV)	0.743	+0.023	14 %
32B free-gen single pass (SOTA bar)	0.819	—	—
oracle@8 (luck ceiling)	0.879	+0.159	100 %

The best training-free selector captures only **24 %** of the gap above random, and *none* beats the 32B's own single pass (0.819); handing the 32B the 7B's candidates to synthesize actually **backfires** (0.774 vs 0.819). The mechanism is a **majority trap**: the correct answer is a minority vote in **74–90 %** of recoverable questions (mean ≈ 1.5 of 8 votes), so any consistency/agreement rule is systematically pointed at the wrong answer. This is exactly the floor the *trained* verifier (§4.7) breaks — and even it is bounded above by selection efficiency.

Per-answer the trained verifier ranks correct above incorrect at AUROC ~ 0.90 (Lingshu 0.903, MVT 0.913, IV3 0.898), yet its **selection efficiency** — how often argmax picks a correct candidate when a recoverable one exists — is only **81 % / 82 % / 74 %**, for the same majority-trap reason. A five-experiment battery establishes the gap is largely **fundamental**, not a fixable artifact:

- Compounding fails.** Diverse-generation (+0.030) and pairwise-selection (+0.021) each beat the pointwise-iid baseline, but do **not** stack: pairwise-over-diverse (0.5376) $\leq$ pointwise-over-diverse (0.5494), and on PMC the diverse +0.110 oracle lift stays unconverted (pairwise converts 0.005). **Diversity buys coverage, not selectability.**
- Distractor-filtering fails.** Eight pre-filters over the diverse pool; the best (rarity_log1p 0.5601) does not beat *both* baselines and sign-flips per dataset. Mechanism: the correct *new* answers diverse generation adds are themselves rare, so a rarity/agreement signal cannot separate correct-rare from wrong-rare.
- Verifier capacity does not break it.** A **32B-zeroshot** verifier (7× capacity) only **ties** the 7B-trained one: sel_acc **0.480 vs 0.475, +0.005, CI [−0.023, +0.032], n.s.** The pure-capacity contrast (32B-zeroshot vs 7B-zeroshot +0.067) is real but small, and the 32B still leaves an oracle → selection gap of 0.192 (conversion 0.15). **The ceiling is substantially fundamental, not a capacity artifact** — a 7× verifier recovers ~ 7 of the ~ 19 oracle-gap points.
- Post-hoc selectors fail.** *Simulated* active-pairwise, bandit allocation (Thompson/UCB-E $\approx$ uniform), and unsupervised Dawid-Skene aggregation ($\approx$ pooled majority, +0.132 below the trained verifier) all collapse to the pointwise/uniform/majority baseline — unsupervised reliability tracks self-agreement (~ 0.52), not accuracy (~ 0.29): the generators are **confidently wrong**.
- The binding limit is therefore **candidate quality** (oracle@8 $\gg$ verifier-bo8 everywhere: pooled Lingshu 0.513 vs 0.414); cross-model pooling lifts oracle **+0.11–0.15** — the largest remaining headroom, but only convertible with a stronger selector (the real pairwise verifier converts part of it).

6.4 The Pareto reframe of best-of-N — dominated by a cheap strong leg, alive against the slow one

Whether best-of-N is *deployable* depends entirely on the strong leg's cost. Against the **cheap no-think 32B** (4.57 FLOP-eq, 665 ms) best-of-N is **FLOPs-dominated**: the global FLOPs-Pareto envelope is greedy → 7B+Pandora → always-32B (7B-greedy $F = 1.00 / 0.518$; iid→Pandora $F = 2.0-3.25 / 0.519-0.568$; always-32B $F = 4.57 / 0.673$), and **diverse-gen is not on the envelope** (iid→Pandora dominates it at every accuracy target), so its 1.875× generation cost never repays. On **latency**, though, best-of-N is **alive**: a parallel best-of-N base is only **522.6 ms = 0.79×** the single 665 ms 32B forward (batch-1 short-gen is overhead-bound, so the 32B is only ~1.9× the 7B wall-clock, not 4.57×) — but it still does **not beat always-32B** (fixed-bo-N tops ~0.55, far below the 32B's 0.673; matching 0.673 forces heavy escalation that pushes parallel latency back above 665 ms). **However**, the deliverable's baseline is not the 665 ms no-think leg — it is the **10 521.6 ms / 2 001.9 J think** leg. Re-priced against *that*, the parallel best-of-N leg is ~0.05× its latency (a ~20× win) and the July-6 "best-of-N is not deployable" verdict is correctly **scoped to the no-think baseline only**. Net conclusion: the deployable efficiency win is the **router** (\$5.7: 48-82 % of always-32B FLOPs at parity) **plus Pandora for tight compute budgets** ($F = 2-3.3$ for acc 0.52-0.57); the best-of-N leg is the **accuracy engine** whose latency dominates the *think* baseline it replaces. (Scope caveat, honest: in an expensive/slow-strong regime the parallel best-of-N leg could win on FLOPs too — that regime is not measured here.)

6.5 Adjacent negatives (kept for the record, not abstention)

- **The action axis is capacity-bound.** Cheap same-model repairs (look-closer / think) recover a real but **unharvestable** 14.3 % of the 7B's errors that the 32B *also* misses (stable 11-17 % across benchmarks), but the repairs break as many answers as they fix (per-view acc noise-limited: cap320 0.622, full-res 0.621, think 0.607 vs 0.645 for the 32B), so a confidence-gated repair ladder *loses* at 32B-parity. The 32B's edge is capacity, not a cheap transform the 7B could apply to itself.
- **Cross-family complementarity is real but unexploitable.** Oracle union(7B | InternVL | Phi) = 0.801 vs always-32B 0.645, but a learned router over frozen peer signals captures none of it (best 0.621 ≈ always-7B 0.622; a SigLIP image+text router is at chance, AUROC 0.50).
- **The cheap model leans on a language prior.** 56.9 % of the 7B's answers are unchanged when the image is blanked, and image-sensitive vs image-insensitive questions have near-equal accuracy (0.620 vs 0.625) — a large fraction of "VQA" carries no routable visual difficulty.

None of these is presented as a method. Each is a *negative* that maps where effort pays: change the structure (ACC) or add a little training (the verifier), because the frozen-model routing/selection signal is, at training-free effort, luck.

7. Reproduction of the 3-family suite + the OmniMed strong-leg fallback

6 of 7 benchmarks (MMMU, VQA-RAD, SLAKE, PathVQA, PMC-VQA, MedXpert): fully run cheap + strong for all three families (+ think tier), faithful vs the Lingshu paper (\$2.3). The 7th, OmniMedVQA (Open-access, **88 996 QA**, 4-option MCQ, 42 sub-datasets of which RadImageNet is ~64 %), is cheap-faithful + strong-fallback:

- **Cheap (7B/8B) legs ran and reproduce the paper** over the full 88 996-QA set: Lingshu-7B **0.8274** (73 639 correct, paper 0.829 — a 0.2-pt reproduction), MedVLThinker-7B 0.6248, InternVL3-8B 0.7847. Per-modality breakdowns are internally sensible (Lingshu-7B Modality-Recognition 0.986, weakest on CT 0.772), confirming the OmniMed pipeline is faithful after the July-3 `modality_type` parser fix.
- **Strong (32B/38B) legs are BLOCKED** by a deterministic **tp = 2 NCCL collective hang** on the 89k-image queue (every chunk stalls ~36 min, the heartbeat watchdog aborts, the driver re-fires and re-hangs; confirmed over ~2 days). Ruled out: chunked **tp = 2 + retry × 3**, `TORCH_NCCL_ENABLE_MONITORING=0` (hangs forever), `EVAL_BATCH_SIZE=256` (fixes the cgroup OOM, not the hang), 3 h/chunk timeout + aggregation backstop. **tp = 1 is not an option** (64 GB weights + multimodal activation OOMs the 80 GB card; the `MAX_MODEL_LEN` KV-cache lever cannot buy it back).
- **Decision (why it is safe, not a cop-out).** OmniMed is a **keep-cheap** benchmark — paper Lingshu-7B 82.9 vs 32B 83.4, a 0.5-pt gap; our cheap 0.827 already matches — so a cascade **keeps-cheap at ~0 % escalation** and the missing strong number changes **no cascade conclusion**. We report the strong OmniMed cell as **paper-reference (Lingshu-32B 0.834) + infra-limited**, back the fidelity claim with the faithful cheap reproduction, and — per the standing rule — write **no fabricated metrics.json**. Net: **6/7 faithful + OmniMed cheap-faithful + strong-fallback**; only the pooled ALL-7 vs-32B-think figure is missing (not any verdict), and **ALL-6 is the computable pool**. (INT4-32B at **tp = 1** ~20 GB would fit one GPU and avoid this hang — a future re-run path, \$3.5.)

Reproduction engineering recorded honestly (the fixes are part of the deliverable): the OmniMed `KeyError` over 42 sub-datasets (`sample.get(...)`), the `MAX_MODEL_LEN` / `GPU_MEM_UTIL` env levers on both vLLM wrappers, the chunked resumable **tp = 2** strong driver, and the strictly-sequential cheap driver that dodged the two-at-a-time cgroup OOM (anon-rss ≈ 245 GB in the kill log).

8. Related Work and Novelty

We state, inline and without hedging, exactly which mechanisms we adopt unchanged and which are new. The short version: several components are established mechanisms transferred into a new setting; the contribution is the *assembly* (a format-aware regime-adaptive router scored against the honest think baseline), the *compute-configuration structure*, the trained-verifier *application and unification*, and the *characterization of the two walls*.

Efficient cascades and query routing. FrugalGPT popularized the cost-saving LLM cascade. Agreement-based cascading (**ABC**, arXiv 2407.02348) escalates on ensemble *disagreement*; **CAR** (confidence-aware routing, arXiv 2505.15154) is the closest structural prior art. Our think-gate is, by construction, an ABC member; the tier-0 margin gate is Chow/MSP. What is new is *what the gate switches between* — three **compute configurations** of the *same two models* (7B-nt $\rightarrow$ 32B-**no-think** $\rightarrow$ 32B-think), inserting the strong model's fast no-think mode as an intermediate tier because reasoning over-thinks perception.

Confidence, selective prediction, conformal deferral. The training-free gate family we benchmark against margin is the classical selective-prediction toolkit (MSP/Chow, entropy, Gini/DOCTOR, split-conformal / CP-Router, learning-to-defer). In our bake-offs these cluster at the recoverability ceiling; none separates from margin (MCQ) or verifier-confidence (open). **We use selective/conformal machinery only to decide which model answers (Wilson-LB certification, held-out τ), never to withhold an answer.**

Post-hoc deferral theory — the wall. Jitkrittum et al. (NeurIPS 2023, arXiv 2307.02764) prove confidence-only deferral is fundamentally limited by how predictable the strong model's *marginal correction* is; their Diff-Prob score is the SOTA post-hoc gate. This is a precise description of our recoverability wall: "will the strong model fix this?" is only AUROC ~ 0.6 from any cheap signal, and a faithful Diff-Prob gate reaches only AUROC 0.708 (MCQ) / 0.744 (open) — below the deployed signals — because the strong model repairs only 6–10 % of cheap errors (26 % where competitive) and *which* is near-unlearnable. We treat Jitkrittum as the *explanation* for the wall, not a baseline to beat.

Verifiers and reward models for best-of-N. Generative verifiers (**GenRM**) recast reward modeling as next-token "is this correct?" prediction for best-of-N selection in text; VLM process-reward models rerank reasoning steps. Our verifier is squarely in this lineage — a small LoRA **outcome** head $s_\phi = P_\phi(\text{Yes} \mid v, q, a)$ with a BCE objective — and we claim no architectural novelty. The contribution is the **application and unification**: a single outcome verifier for inference-time selection in **medical VQA**, trained and evaluated over **both free-text answers and bounding-box grounding** ($y = 1[\text{IoU} \geq 0.3]$). We position it as a constructive counter to **Verification-Mirage** (arXiv 2605.10850), which concluded self-verification *fails* in medical VQA: that result is about *zero-shot* self-verification (our evidence agrees: zero-shot self-verify $\approx$ AUROC 0.5, a zero-shot 32B verifier 0.355 < a trained 7B 0.403), and the difference is **training** (the trained head reaches AUROC 0.924).

Verifier-score-as-gate precedents. **CCPS** (arXiv 2505.21772) scores reliability via input-perturbation stability; **Self-REF** (arXiv 2410.13284) trains self-reference confidence tokens; conformal treatments of verifier scores exist (Kiyani et al. arXiv 2602.17633). Our escalation gate — thresholding the trained verifier's confidence — is in this spirit and we claim no novelty for the *idea*; where these overlap our setting we benchmark them faithfully (a CCPS-style visual-stability gate reaches only AUROC ~ 0.60 and adds nothing to verifier-confidence).

Cross-field transfers (named, so the borrowing is explicit). The adaptive open-arm controller is **Weitzman's Pandora's-Box** optimal-search rule (Econometrica 1979), mapped so each "box" is a cheap draw or a 32B escalation and one exchange-rate knob λ yields both thresholds. The candidate-pooling analysis is **Markowitz** portfolio theory (asset = generator, covariance = error- ϕ); the unsupervised aggregation baseline is **Dawid-Skene**; the fusion analysis reduces to **Chair-Varshney** decision fusion; the pairwise selector is a **PairJudge-RM / knockout-tournament**; the credibility audit is **Bühlmann-Straub**. The visual-token-prune lever is **FastV**-style shallow-exit. Each is imported as a testable variation and reported with its verdict (most are honest negatives or regime-limited refinements).

Base families and the faithful protocol. Lingshu-7B/32B (arXiv 2506.07044), MedVLThinker-7B/32B, InternVL3-8B/38B; verifier base Lingshu-7B, box-verifier base Qwen2.5-VL-7B. We reproduce the Lingshu MedEvalKit protocol faithfully (§2.3) rather than through an internal harness.

Open-ended vs multiple-choice routing — the discreteness claim. The observation, which we did not find stated elsewhere, is that the *rouatability* of a medical VLM is largely an MCQ artifact: the same confidence signal moves from AUROC ~ 0.6 (MCQ) to ~ 0.87 (open-ended), driven by answer *discreteness*, not *length* (token-F1 agrees). The corollary — study medical-VLM cascades and verifiers open-ended — motivates the verifier results.

9. Limitations and honest caveats

- **Fusion / best-of-N cost FLOPs.** The PMC fusion cell runs both legs (+22 % FLOPs on that slice); the open-text best-of-8 arm is 16 cheap forwards (break-even vs one 32B forward is $N \leq 2$). These are **latency + accuracy** levers, not FLOP-savers. The *MCQ margin cascade* is the FLOP-saving part (1.74 pooled MCQ FLOPs vs 4.57). F8 (V2) makes the accuracy-max PMC cell FLOP-negative but only *captures 70 %* of F3's PMC beat.

- **Open-text 32B-THINK accuracy is ESTIMATED** for the open cells (judged 32B-no-think + the measured modal think-delta $-0.195 / -0.120 / -0.130$). A judged 32B-think open-text dump would remove the estimate. PathVQA-closed has no 32B-think dump → its think acc is set = no-think.
- **In-domain verifier.** SLAKE-open / VQA-RAD-open / PathVQA-open are **in-domain** for the pooled4 verifier (trained on slake + pathvqa + kvasir + vqa_rad); the open-text pooled-4 numbers are in-domain, not held-out-domain (RadImageNet is the held-out-domain transfer, +13.6 % of its gap). The from-scratch other-base (MedVLThinker) verifier is not uniformly robust (fails VQA-RAD at $n = 54$). Against the strong model the verifier is matches-to-modest-win (mean +0.017; seed-0 significant, seed-1 a tie), robust only over training-free selection; it lifts *selection over a frozen generator*, it is **not** a SOTA grounder (absolute box IoU stays modest: SLAKE 0.255, MS-CXR 0.232).
- **OmniMed-32B blocked** (deterministic tp = 2 NCCL hang) → ALL-7 vs-32B-think is unreportable; ALL-6 is the computable pool. Conclusion unchanged (keep-cheap).
- **MMMU is $n = 150$ and a Lingshu-7B-specific anomaly** (+26 over paper), excluded from all *fidelity* claims; the router *exploits* it as a route-to-7B override (+0.140 vs think) but we never claim it as a fusion win, and its small n means its macro contribution should be read with the CI in mind (keep-7B d = +0.167, CI [0.087, 0.247]).
- **INT4 (G3) and G4 are PROJECTED and labelled as such.** INT4 latency is a composition-grounded projection (prefill-bound leg → 665 → 583 ms, not 2.5×); INT4 FLOPs are unchanged under MAC accounting; INT4 accuracy is a literature bracket. Empirical AWQ latency + Lingshu-32B-INT4 accuracy are future work (no loadable AWQ Lingshu-32B; CDN outage stalled the committed benchmark).
- **Judged-think is estimated; two eval contexts are kept separate.** Do not cross-multiply the faithful MedEvalKit eval (accuracy + reasoning think-deltas) with the 5-family NGC ACC bake-off (measured batch-1 latency/energy). Some June/July latency cells (InternVL3 family, PathVQA rows) were measured under GPU contention and are uncitable pending a serial re-run (accuracy and FLOPs unaffected). G8's $\phi = 0.586$ is transferred from an MVT-32B measurement as a *fraction*; a direct Lingshu-32B prefill/decode split would remove the transfer, and unconditional-prefetch FLOPs assume an idle 2nd GPU.
- **A minor within-repo inconsistency, flagged:** `escalation_levers.py` / `beat32b_fusion.py` score SLAKE-open with the greedy+seqlogprob FALC fallback (esc ~53 %, 698.6 ms — a "slower cell"), whereas the final method (`integrated_method.py`) uses bo8+verifier SLAKE-open (605.5 ms, esc 12.6 %), so those artifacts' pooled baselines differ slightly (0.5749 / 2.337 vs 0.5750 / 2.538). The final method is the bo8+verifier treatment.

10. Conclusion

A medical-VQA cascade must decide **which** queries to escalate and **what** to run when it does. Scored against the honest expensive baseline it is actually about — **always-32B-THINK** (measured 10.5 s / 2.0 kJ per open-text question) — this paper's **format-aware, regime-adaptive router** is *faster and more accurate*: on the full MedEvalKit suite ($n = 42\,374$) the compute-lean setting is **+0.0123 vs think at 0.49× its FLOPs and −96 % latency**, and the accuracy-max V2 setting is **+0.0212 at 0.93× FLOPs** — all-axes dominant. Always-32B-THINK is Pareto-dominated on all five families, because thinking over-thinks perception VQA. The two positive levers are **structure** (the ACC compute-configuration cascade: −80 % latency, FLOPs → 52 % at ALL-6 parity) and **a little training** (the outcome verifier: beats the 32B/38B across three families and held-out OOD, +0.191 on real MS-CXR boxes). Both are honestly bounded by two **walls** we present as a headline: the **recoverability wall** confines the broad-slice MCQ *beat* to PMC-VQA (six independent methods + an FDR-controlled slice search agree, and the CI-guardrail is the correct fix per an H8 credibility audit), and the **selectability wall** makes best-of-N's oracle → selection gap largely fundamental (a 7× verifier only ties the trained 7B; coverage does not compound with selection). The deployable efficiency win is the router + Pandora; the trained verifier is the accuracy engine whose latency dominates the think baseline it replaces. We release the unified pipeline (`method_final.py`, one command, CPU re-costing of saved dumps), the trained answer/box verifiers, and the full negative-result characterization so both the wins and the walls reproduce end-to-end. The clearest direction for future work is the binding limit throughout — **candidate quality** — where a cross-model pool raises oracle@N by +0.11–0.15, the largest remaining headroom.

Reproducibility index

Every number traces to a checkpoint; code launches from the repo root (`~/medvlthinker-imgdiff-compute`).

The unified method (one command). `python3 src/cascade_methods/method_final.py` → `artifacts/method_final.json` (V1: both knob settings + INT4 + reconciliation) **and** `method_final_v2.json` (V2: F8 + F10). Per-lever references: `integrated_method.py` (`integrated_method_vs_think.json`), `beat32b_fusion.py` (`beat32b_fusion.json`), `beat32b_more.py` (`beat32b_more.json`: F7/F8/F11/F10), `integrated_pandora.py` (`integrated_pandora_opentext.json`), `pandora_controller.py` (`pandora_controller.json`), `quantized_strong_leg.py` (`quantized_strong_leg.json`), `escalation_levers.py` (`escalation_levers.json`), `escalation_more.py` (`escalation_more.json`), `logit_fusion.py` (`logit_fusion.json`),

robust_slice_routing.py (robust_slice_routing.json: H4 discovery + H8 credibility), unified_router.py (unified_router.json).

Baseline costs (measured batch-1, NVML). opentext_32b_think.json (32B-think open-text 10 521.6 ms / 2 001.9 J), iv3_38b_latency.json (IV3-38B 6 220 ms / 3 275.6 J), latency_32b.jsonl (prefill/decode composition, $\phi = 0.586$), reframe_vs_bigthink.json (5-family ACC reframe), master_data.csv (ACC ALL-6/ALL-5).

ACC (structure / efficiency). src/cascade_methods/{acc.py, acc_v2.py, acc_v3_confgate.py, acc_v4_lowres_think.py}; over-thinking premise {strong_leg.py, final_comparison.py, overthink_generalize.py}; bake-off / cross-family {acc_allmethods.py, acc_compare.py, cascade_all_families.py, gate_compare.py}; per-tier batch-1 latency/energy {latency_estimate.py, open_measure_latency_energy.py} + src/labeling/nvml_power.py; cost math METHOD_ACC.md / METHOD_MATH.md.

The two walls (the negatives). Recoverability: beat32b_fusion.json, beat32b_more.json, logit_fusion.json, robust_slice_routing.json; luck-floor / recoverability diagnostics src/cascade_methods/{diagnostics.py, strong_fixes_genuinely_unknown.py, knowledge_feasibility_bytype.py}; the gate bake-off gate_unified_bakeoff.py (gate_unified_bakeoff.json). Selectability: {combine_diverse_pairwise.py, distractor_filter.py, verifier_32b_measure.py, end_to_end_consolidation.py, latency_reexamination.py, active_comparison_verifier.py, bandit_allocation.py, dawid_skene_aggregate.py, generator_portfolio.py, diversity_candidates.py, diverse_measure_gpu.py, pairwise_verifier_score.py, pandora_correlated.py, pandora_pooling_combo.py} with matching artifacts.

Trained verifier (accuracy). Answer verifier src/training_methods/{run_lora_verifier_open.py, run_lora_verifier_ranking.py, clean_verifier_dump.py}; non-circularity / active-ingredient ablations {verifier_image_ablation.py, verifier_transfer_eval.py, verifier_scaling_curve.py, cross_gen_verifier.py}; 3-family beats-32B table src/cascade_methods/open_verifier_cascade_table.py; gate bake-off / cost frontier {open_gate_swap.py, open_gate_efficiency.py, open_gate_heldout_tau.py, open_recoverability_gate.py, open_cost_frontier.py}; box verifier src/training_methods/run_lora_box_verifier.py; grounding src/labeling/{run_ground_slake.py, run_ground_mscxr.py}; UGV MCQ src/cascade_methods/ugv_mcq_verdict.py (ugv_mcq_verdict.json).

Open-ended ceiling-break + judge. src/labeling/{run_openvqa.py, run_openvqa_fewshot.py}; gate hunt src/cascade_methods/{gate_search_open.py, open_gate_bakeoff.py}; judge src/labeling/run_judge.py (MedVLThinker-32B + Claude-Sonnet-5 cross-check, κ 0.85-0.96, 100 % exact-match anchor).

Docs (canonical ledgers). results/cascade_methods/docs/current/{METHOD_FINAL_2026-07, RESEARCH_RESULTS_2026-07, METHODS_MASTER, MASTER_SUMMARY_2026-07, OMNIMED_FALLBACK, METHOD_ACC, METHOD_MATH}.md; backlog METHOD_IDEAS_BACKLOG.md (56 ideas); session narrative progress/progress_{June_17-June_29-30, July_01-02, July_03, July_04, July_05, July_06, July_07}.md.

End of manuscript. Every figure above is verbatim from a real checkpoint or arithmetically derived from one; none is fabricated. This document supersedes no experimental data — it consolidates it.